Crash insurance and the G10 carry premium

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Abstract

Does the price of crash insurance in G10 currency options account for the carry premium, and for its change when interest rates rose from 2022? Under a pre-registered design, I price one-month 10Δ protection on a carry portfolio from smiles calibrated to dealer quotes for nine currencies over 2013–2026. The skew price of protection per unit of forward discount falls from 0.81 to 0.45 between the zero-rate and hiking regimes, insignificantly: the forward-discount spread widened while the price of insurance did not change. The ratio predicts carry returns in sample but not out of sample. Truncated quotes identify the sign of risk-neutral skewness only under tail assumptions close to the strongest they admit.

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1 Introduction

A currency carry trade funds positions in high-yield currencies by borrowing in low-yield ones. Its losses are concentrated in bad states. Brunnermeier, Nagel and Pedersen (2008) show that high interest-rate differentials predict negatively skewed currency returns, that carry losses coincide with rises in the VIX and with the unwinding of speculators' positions, and that risk reversals rise after carry losses. If the carry premium compensates for crash risk, the price of insurance against crashes, which foreign-exchange option smiles quote directly, should account for part of it. Burnside, Eichenbaum, Kleshchelski and Rebelo (2011) hedge carry positions with options close to the money and characterise the peso state as one in which the stochastic discount factor is high. Jurek (2014) forms crash-neutral carry portfolios with out-of-the-money options and finds that crash premia account for at most one third of carry returns. Farhi, Fraiberger, Gabaix, Rancière and Verdelhan (2015) estimate the share of the carry premium that compensates for disaster risk. The relation between interest differentials and currency returns has also been unstable: Bussière, Chinn, Ferrara and Heipertz (2022) find positive Fama-regression coefficients from about 2006 through the decade after the financial crisis, Engel, Kazakova, Wang and Xiang (2022) find that, for the dollar, interest differentials stop predicting excess returns when rates are near the zero lower bound, and Andrews, Colacito, Croce and Gavazzoni (2024) relate the lower return to traditional carry after 2008 to lower expected global growth and inflation.

This paper asks whether the price of crash insurance, measured from G10 foreign-exchange option smiles, accounts for the carry premium and for its change between the zero-rate regime (May 2013 to December 2021) and the hiking regime (from January 2022). The regime labels follow the design and denote calendar periods, not rate paths. The design's question also names the collapse of the carry premium after 2008. The price of 10Δ protection cannot be measured before the one-month risk-reversal quotes begin, in 2007 for Fenics and between 2010 and 2013 for the composite, so the estimands compare the zero-rate regime from 2013 with the hiking regime, and the collapse appears only in the realised returns of the long at-the-money sample (Section 5.3): unhedged carry earned 61.4 bp per month over 1995–2007 ($t = 3.17$), 33.1 bp over 2009–2021 ($t = 1.95$) and 37.0 bp from 2022 ($t = 2.08$). The ex-ante carry of a three-long,

three-short portfolio, its forward-discount spread, averages 18.61 basis points (bp) per month in the first regime and 27.05 in the second (a supplementary, post hoc split). Realised carry returns, compared by regime in a further post hoc analysis, average 7.4 bp per month in the first regime and 37.0 in the second, but the difference (29.6 bp, standard error 23.8) is not significant, and their full-sample mean is itself insignificant ($t = 1.47$). The regime comparison therefore concerns the forward-discount spread, and asks whether the option market's price of left-tail protection moved with it.

I fixed the research design on 23 September 2026, before any return, hedge cost or option-implied statistic was computed.[1] Later changes are recorded in the research log with their date and reason, and where a result had been computed under the original rule, it remains reported. Month-end smiles for nine currencies against the dollar are calibrated to dealer-contributed at-the-money, 25Δ risk-reversal and 25Δ butterfly quotes from LSEG Workspace. The carry portfolio is formed as in Lustig, Roussanov and Verdelhan (2011), and each leg is protected by a one-month 10Δ option that pays when the leg loses. The primary estimand (E1), φ, is the skew premium of that protection (the premium at the smile volatility less that at the at-the-money volatility, same strike) per unit of the portfolio's forward discount. E2 asks whether φ predicts carry returns; E3 splits the realised hedge cost into payoff (delta and diffusive), volatility-level and skew terms; E4 repeats E1 and the skew term under the other butterfly reading; E5 measures exposure to VIX-futures roll-down and to global FX volatility (Menkhoff, Sarno, Schmeling and Schrimpf, 2012).

The primary estimate does not support a claim. Under the market-strangle reading with 10Δ hedges, φ averages 0.809 in the zero-rate regime and 0.454 in the hiking regime; the difference of -0.355 has a Newey–West standard error of 0.204 and a stationary-bootstrap 95% interval of $[-0.814, 0.014]$, so I do not claim a regime difference. A supplementary split, added after the first results, shows why the point estimate falls: the skew price of protection is almost unchanged, at 11.24 bp per month in the first regime and 11.19 in the second, while the forward-discount spread rises by 8.44 bp (Figure 1). After the small-sample bias correction (Stambaugh, 1999), a higher φ predicts higher carry returns in sample (one-sided $p = 0.021$), the sign a crash-compensation reading implies, but the out-of-sample Clark–West statistic of 1.639 does not reach the 5% critical value of 1.645, and about four fifths of its numerator comes from forecasts of returns in 2020 (post hoc; Figure 2). Only one of the three pre-registered conditions for explaining the regime shift is met, and it concerns in-sample predictability rather than the change between regimes, so the explanation has partial support only.

The ex-post evidence is consistent with this reading. Over 2013–2026 the mean realised cost of 10Δ protection is close to the skew premium paid ex ante (-11.2 bp per month), because the payoff and volatility-level terms are indistinguishable from zero. The hedge-cost ratio θ_{UB} is 0.53, against a diffusive null of about 0.10, but its 95% test-inversion set (Fieller, 1954) is unbounded and I claim no value for it. A longer sample from 1995 with at-the-money protection bounds the set, but the bounded set contains the diffusive null of an at-the-money hedge, and the one-month risk-reversal quotes that an out-of-the-money hedge needs start only in 2007. Unhedged carry loads strongly on the VIX roll-down; the hedge lowers the loading by about a fifth, and neither portfolio earns a significant α. Among the robustness variants, only a ten-currency ranking in which the dollar can be a leg makes the difference in φ significant under both the Newey–West and the bootstrap inference, and no variant rejects out of sample when training starts in 2013, a training start that departs from the design's rule for E2 (Section 4.7). The estimates point to a rise in the forward-discount spread rather than a change in the price of crash insurance.

Section 2 sets out what option prices can and cannot identify. It restates the decomposition of the currency premium of Backus, Foresi and Telmer (2001), with the conditional entropy of

[1] Pre-registered here means fixed in the project repository (`research_design.md`, dated 23 September 2026) on the day the data were retrieved and before any return, hedge cost or option-implied statistic was computed. The design was not lodged with an independent registry, and the regime boundary was chosen with public knowledge of the path of interest rates and exchange rates.

Backus, Chernov and Zin (2014) and with the covered-interest-parity basis as a wedge, much as Lustig and Verdelhan (2019) do with a general wedge: option prices identify the risk-neutral law of the log exchange-rate change but not, without a parametric model of the discount factors, how the entropy difference behind the currency premium divides across cumulant orders, so the design measures the price of the carry position's left-tail exposure, not the crash share of discount-factor entropy. I also bound option-implied moments from truncated quotes, in the spirit of the truncation bounds of Jiang and Tian (2005) but under a stated tail condition, and derive the diffusive null of the hedge cost, whose leading-order form is in Jurek (2014) and Farhi et al. (2015), in a mean-value form with an explicit error bound. For the tail condition I show that the exponents of the power-law tails that match the boundary price and slope (Benaim, Dodgson and Kainth, 2008; Qin, Yang, Che and Feng, 2026) are the largest the whole class admits and that the tail is then unique in it, derive the sharp identified set of the moments, which shows that, for nonnegative weights, the upper bounds of R4 are attained and, for weights positive near the boundary, its lower bounds are not, and give a check of the boundary-elasticity setting and a breakdown point for the sign of skewness (R10).

The closest work is Jurek (2014), who studies unhedged carry over 1990–2012 and option-hedged carry over 1999–2012, and Daniel, Hodrick and Lu (2017), who hedge G10 carry with 10Δ and 25Δ options over 2000–2013 and find that hedging reduces but does not eliminate its abnormal returns; this paper extends option-hedged evidence for G10 carry through the hiking regime to 2026. Jurek and Xu (2014), according to their abstract, calibrate a non-Gaussian model of pricing-kernel dynamics to G10 option prices and attribute about 15% of the *HML* premium to skewness and higher moments of the kernels; that division rests on their parametric model, whereas this paper measures the option market's price of 10Δ protection on the carry portfolio and its change between regimes. Burnside et al. (2011) hedge with options close to the money; the hedged return here follows their construction, but the primary hedge is at 10Δ, because a one-month 25Δ option is only about 2% out of the money. Farhi et al. (2015) estimate a disaster share and, like Jurek (2014), state the leading-order form of the diffusive null (Section 2.4); this paper estimates prices and exposures, not a disaster share. Chernov, Graveline and Zviadadze (2018) estimate the jump share of the conditional entropy of dollar exchange-rate changes over 1986–2010, which is distinct from the share of the carry premium. Della Corte, Ramadorai and Sarno (2016) use the volatility risk premium as a currency predictor, read the quoted butterfly as a smile strangle and sort currencies on the 10Δ risk reversal; here the effect of the reading is measured (E4). Li, Sarno and Zinna (2026) build a skewness risk premium for each currency from over-the-counter options and find it priced in the cross-section of currency returns; here the object is the time series of the skew price of protecting a carry portfolio and its change between regimes. Fan, Londono and Xiao (2022) show that equity tail risk, measured from one-month out-of-the-money S&P 500 puts, is priced in currency returns; here the tail price comes from the currencies' own options. Caballero and Doyle (2012) relate carry returns to a VIX futures roll-down strategy, which E5 adapts to month-ends, and argue that exchange-rate options are cheap systemic insurance.[2]

The paper adds three pieces of pre-registered evidence to this literature. The first is a decomposition of the realised cost of G10 crash insurance over 2013–2026 into payoff, volatility-level and skew terms, in the pooled sample and in each of the two rate regimes. The second is a pre-registered test of whether the ex-ante skew price explains the change in the carry premium between the regimes. The third, secondary, is the size of the difference between the two readings of the butterfly quote for the carry estimands; Reiswich and Wystup (2012) set out the market-strangle calibration on which the distinction rests, and Bossens, Rayée, Skantzos and Deelstra (2010, Section 3.3) show that the two readings can differ markedly when the smile is

[2]Choi and Suh (2022), according to their abstract and introduction, use risk reversals to predict carry crashes and to hedge conditionally; the hedge here is unconditional. Kutuk and van Wijnbergen (2025) apply the hedged-versus-unhedged comparison to USD/TRY only.

skewed. Two further contributions come from post hoc analyses and are labelled as such. One is a direct comparison of the regime changes in the skew price and in the forward-discount spread, which bounds the share of the rise in the spread matched by the price of crash insurance and, read through the model of Farhi and Gabaix (2016), locates the rise outside disaster exposure. The other is methodological. Bondarenko, Dillschneider, Schneider and Trojani (2026), according to their abstract, show that under market incompleteness the valuation uncertainty that an observed option cross-section leaves about essentially all standard risk-neutral variance, skewness and kurtosis measures is unbounded. R10 asks instead how strong a stated tail restriction must be for the sign of risk-neutral skewness to be identified when quotes stop at the 10Δ strikes, measures it in G10 data and in Merton (1976) and Heston (1993) models, and tests it out of sample with held-out 5Δ quotes.

Section 2 presents the framework, Sections 3 and 4 the data and method, Section 5 the results and Section 6 their robustness, and Section 7 concludes. Appendix A lists the theory results, Appendix B gives the steps needed to reproduce the results, including a route that recomputes the portfolio-level results from published month-level series without an LSEG licence, and Appendix C records every departure from the design.

2 Framework: what option prices identify

The results are labelled R1 to R10 as in `theory/notes.tex`, which contains the proofs of all of them except R9, which is cited; they are stated here with their hypotheses. Appendix A lists their status and sources. Throughout, Φ is the standard normal distribution function, τ is the time to expiry and, for forward F, strike K and volatility σ, $d_\pm(K,\sigma) = [\ln(F/K) \pm \sigma^2\tau/2]/(\sigma\sqrt{\tau})$; with $s = \sigma\sqrt{\tau}$ fixed this is written $d_\pm(K)$.

2.1 Currency premia and discount-factor entropy

Let S_t be the USD price of one unit of a foreign currency, $s_t = \log S_t$, f_t the log one-period forward, r_t and r_t^* the one-period USD and foreign interest rates, and

$$x_t = (f_t - s_t) - (r_t - r_t^*) \tag{1}$$

the covered-interest-parity basis of Du, Tepper and Verdelhan (2018).[3] For a positive random variable M with $\mathbb{E}_t|\log M| < \infty$, conditional entropy is $L_t(M) = \log \mathbb{E}_t M - \mathbb{E}_t \log M \geq 0$.

R7 (entropy decomposition). *Let $M, M^* > 0$ be $\mathcal{F}_{t+1}$-measurable discount factors in L^2 that price USD and foreign payoffs in their own currencies, with $\mathbb{E}_t M = e^{-r_t}$, $\mathbb{E}_t M^* = e^{-r_t^*}$, $\mathbb{E}_t|\log M|, \mathbb{E}_t|\log M^*| < \infty$ and $MS_{t+1}/S_t \in L^2$. Suppose $M^* = MS_{t+1}/S_t$, which holds if every foreign payoff in $L^2(\mathcal{F}_{t+1})$ is traded and priced consistently in both currencies. Write $m = \log M$ and $\Delta s = s_{t+1} - s_t$. Then:*

(i) $\mathbb{E}_t[\Delta s + r_t^* - r_t] = L_t(M) - L_t(M^*)$ *and* $\mathbb{E}_t[s_{t+1} - f_t] = L_t(M) - L_t(M^*) - x_t$*;*

(ii) if the conditional cumulant generating function of m has a Maclaurin series with radius of convergence greater than one, $L_t(M) = \sum_{j\geq 2} \kappa_{j,t}/j!$, where $\kappa_{j,t}$ are the conditional cumulants of m;

(iii) for any gross return $R > 0$ with $\mathbb{E}_t[MR] = 1$ and $\mathbb{E}_t|\log R| < \infty$, $L_t(M) \geq \mathbb{E}_t \log R - r_t$.

The decomposition is due to Backus, Foresi and Telmer (2001).[4] The name conditional entropy and the bound in (iii) are as in Backus, Chernov and Zin (2014), who relate the bound

[3] They quote exchange rates in foreign currency per USD, so x_t is minus their basis.

[4] With the exchange rate equal to the ratio of the two kernels (their Proposition 1), the expected excess return on a forward position is the difference between the log of the expectation and the expectation of the log of the kernels (their eq. (11)), and higher cumulants enter only through that difference (eqs. (12)–(13)).

to Bansal and Lehmann (1997) and Alvarez and Jermann (2005). Lustig and Verdelhan (2019, Section 2.5) restate the decomposition with a wedge between the exchange rate and the ratio of kernels, and Jiang, Krishnamurthy and Lustig (2021) link the value of the dollar to a convenience yield inferred from the Treasury basis, a covered-parity deviation measured with government bond yields. Writing the covered-interest-parity basis of equation (1) into the identity is an elementary step, derived in the notes.

By (i), the expected log excess return on a foreign currency is the difference of two conditional entropies less the basis. By (ii), entropy is a sum of cumulants of log marginal utility, so crash risk enters through cumulants of order three and above. By (iii), applied to the gross returns of fully collateralised hedged and unhedged carry positions, the two positions give two lower bounds on the same $L_t(M)$.

2.2 Change of numeraire, arbitrage-free prices and spanning

Options are quoted on currency pairs, five of which have the dollar as the base currency. The change of numeraire connects the two directions of quotation.

R1 (domestic and foreign risk-neutral measures). *Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in[0,T]}, \mathbb{Q}^d)$ satisfy the usual conditions, let B^d, B^f be strictly positive money-market accounts with $B^d_0 = B^f_0 = 1$, let $S > 0$ be adapted and càdlàg, and suppose $Z_t = S_t B^f_t/(S_0 B^d_t)$ is a true $\mathbb{Q}^d$-martingale. Define $d\mathbb{Q}^f/d\mathbb{Q}^d = Z_T$. Then (a) $\mathbb{Q}^f \sim \mathbb{Q}^d$; (b) $\mathbb{E}^{\mathbb{Q}^f}[X \mid \mathcal{F}_t] = \mathbb{E}^{\mathbb{Q}^d}[Z_T X \mid \mathcal{F}_t]/Z_t$ for $\mathcal{F}_T$-measurable $X \geq 0$ or $X \in L^1(\mathbb{Q}^f)$; (c) foreign prices discounted by B^f are $\mathbb{Q}^f$-martingales exactly when their domestic values discounted by B^d are $\mathbb{Q}^d$-martingales; (d) under Garman–Kohlhagen dynamics, $\mathbb{E}^{\mathbb{Q}^d} S_T = F$ and $\mathbb{E}^{\mathbb{Q}^f}(1/S_T) = 1/F$.*

This is the change of numeraire of Geman, El Karoui and Rochet (1995), proved in the notes for completeness.[5] In the moment calculations of Section 5.2, prices on a pair whose quote currency is not the dollar are converted to the USD forward measure with density S_T/F, where S is that pair's price of one dollar.

R2 (arbitrage-free call prices). *Let $c : [0,\infty) \to \mathbb{R}$ and $F > 0$. There is a random variable $X \geq 0$ with $\mathbb{E}X = F$ and $c(K) = \mathbb{E}(X-K)^+$ for all $K \geq 0$ if and only if c is convex, $c(0) = F$, $c'_+(0) \geq -1$ and $c(K) \to 0$ as $K \to \infty$. The law of X is then unique, with distribution function $1 + c'_+$ on $[0,\infty)$.*

The link between the second strike derivative of call prices and state prices is due to Breeden and Litzenberger (1978); the characterisation is a known result, proved by the same argument in Föllmer and Schied (2004, Lemma 7.23), and is restated in the notes. On a strike grid R2 implies slopes in $[-1, 0]$ that are nondecreasing, which is the static-arbitrage test used below, together with the density condition on total implied variance of Gatheral and Jacquier (2014); for finitely many strikes at one maturity, Davis and Hobson (2007, Theorem 3.1) give necessary and sufficient conditions of this form, with the zero-strike call included and prices strictly decreasing while positive. An arbitrage-free smile at one maturity therefore identifies the whole risk-neutral law of the terminal exchange rate. The quotes, however, pin the smile down only at the at-the-money, 25Δ and 10Δ strikes.

R3 (spanning). *Let $f : (0,\infty) \to \mathbb{R}$ have a locally absolutely continuous derivative, let $X > 0$ almost surely with $\mathbb{E}X = F < \infty$, and write $P(K) = \mathbb{E}(K-X)^+$ and $C(K) = \mathbb{E}(X-K)^+$. If $\int_0^F |f''| P\, dK + \int_F^\infty |f''| C\, dK < \infty$, then $f(X) \in L^1$ and*

$$\mathbb{E}f(X) = f(F) + \int_0^F f''(K) P(K)\, dK + \int_F^\infty f''(K) C(K)\, dK.$$

[5] If Z is only a strict local martingale, Z_T does not define an equivalent measure; Carr, Fisher and Ruf (2014) construct a foreign measure in that case that is not equivalent to $\mathbb{Q}^d$.

The spanning formula is that of Carr and Madan (2001); the notes write out the standard Taylor-remainder proof with explicit integrability conditions under which expectations can be taken term by term. The volatility, cubic and quartic contracts of Bakshi, Kapadia and Madan (2003) are of this type. They expand around the spot price; the contracts used here are centred at the forward.

2.3 Partial identification from truncated quotes

Quotes observe P and C only between the 10Δ strikes $K_{\min} < F < K_{\max}$, so the tails of the spanning integrals must be bounded.

R4 (bounds on the unobserved tails). *Let $G(u) = \mathbb{Q}(X \le u)$. (a) If $G(u)/u^{\gamma}$ is nondecreasing on $(0, K_{\min}]$ for some $\gamma > 0$, then $P(K) \le P(K_{\min})(K/K_{\min})^{\gamma+1}$ for $0 < K \le K_{\min}$. (b) If $u^{\eta}\mathbb{Q}(X > u)$ is nonincreasing on $[K_{\max}, \infty)$ for some $\eta > 1$, then $C(K) \le C(K_{\max})(K/K_{\max})^{1-\eta}$ for $K \ge K_{\max}$. Hence, for any measurable weight $w \ge 0$,*

$$\int_0^{K_{\min}} w(K)P(K)\,dK \le \frac{P(K_{\min})}{K_{\min}^{\gamma+1}} \int_0^{K_{\min}} w(K)K^{\gamma+1}dK,$$

and similarly in the upper tail. In particular,

$$\int_0^{K_{\min}} \frac{P(K)}{K^2}\,dK \le \frac{P(K_{\min})}{\gamma K_{\min}}, \qquad \int_{K_{\max}}^{\infty} \frac{C(K)}{K^2}\,dK \le \frac{C(K_{\max})}{\eta K_{\max}}.$$

The bounds use only observed wing prices and assumed tail exponents. Sign-changing weights are split into positive and negative parts. Because γ and η are not observed, risk-neutral variance and skewness are partially identified: they are reported as intervals over stated exponents, never as point values. Jiang and Tian (2005, Proposition 2) bound the truncation error of model-free implied variance beyond the quoted strikes; the bounds here rest on a different, stated tail condition and extend to the higher moment contracts.[6] The lemma is elementary, an instance of the monotone form of l'Hôpital's rule, and with an exact power-law tail its bounds hold with equality, which is, in the upper tail, the relative pricing rule of Taleb, Yarckin, Mann, Delic and Spitznagel (2023, Section II-A, eq. (3)) and, in the lower, the power-law put extrapolation of Benaim, Dodgson and Kainth (2008). What I claim is the inequality under the weaker monotone-ratio hypotheses and its use to bound the tails of the moment contracts; I have not found these in this form in the sources I consulted. The proofs are in the notes.

R4 bounds each contract separately and leaves open which exponents the quotes allow. Write $H_L(\gamma)$ and $H_U(\eta)$ for its two hypotheses, and $P_0, G_0, C_0, \bar{G}_0$ for the smile's put and call prices and their strike slopes at the boundary strikes, taken from inside the quoted range ($G_0 = P'(K_{\min}+)$, $\bar{G}_0 = -C'(K_{\max}-)$), which the smile determines.

R10 (admissible exponents and sharp identification). *(a) Some law compatible with (P_0, G_0) satisfies $H_L(\gamma)$ if and only if $\gamma \le \bar{\gamma} = K_{\min}G_0/P_0 - 1$, the strike elasticity of the put price less one, and some law compatible with $(C_0, \bar{G}_0)$ satisfies $H_U(\eta)$ if and only if $1 < \eta \le \bar{\eta} = 1 + K_{\max}\bar{G}_0/C_0$; at $\bar{\gamma}$ and $\bar{\eta}$ the tails are unique power laws and the moments are point-identified. (b) The identified set of the raw moments is the sum of the observed middle part and two compact convex tail sets, each the image of the probability measures μ with $G(K_{\min}z) = G_0 z^{\gamma}\mu([0, z])$ (and the upper-tail counterpart) under one linear constraint; without further price constraints, its extreme points come from measures with at most two atoms. (c) For a nonnegative weight, R4's upper bound is attained, and its lower bound of zero is not sharp when the weight is positive on a set of positive measure in $(K_{\min}z^*, K_{\min})$, where $z^* < 1$ depends on the boundary price and slope*

[6] Lee (2004a) relates the number of finite moments to the limit superior of the slope of implied variance in each wing, the asymptotic counterpart of the tail exponents used here.

and on γ; likewise in the upper tail. (d) If the density is continuous and positive at $K_{\min}$, $H_L(\gamma)$ implies $\gamma \le \gamma_i$, the boundary elasticity of setting (i); setting (i) holds exactly when the reversed hazard rate of $\ln X$ is smallest at the boundary, and it requires $\gamma_i \le \bar{\gamma}$, which says that $\ln P$ is locally concave in $\ln K$ at $K_{\min}$; likewise in the upper tail. (e) The identified sets shrink as the fraction θ of the maximal exponents rises, so, without further price constraints, the fractions at which the sign of the skewness is identified are either none or an interval with right end 1, whose infimum θ^* is the breakdown point of the sign. With further prices as constraints, as in the 5Δ test of Section 5.2, nesting still holds but the right end is the largest fraction consistent with them, and the breakdown point is defined by the computation the notes describe.

The hypotheses are tail-restricted forms of the star order (Lando, Arab and Oliveira, 2023, Definition 2) against the power-function law and the Pareto law; for these reference laws the star order coincides with the reversed hazard rate order and the hazard rate order respectively, a one-line check of mine, and with a density $H_L(\gamma)$ bounds the elasticity of the distribution function studied by Veres-Ferrer and Pavía (2022). The exponent KP'/P at the last strike is proposed as a natural extrapolation exponent by Benaim, Dodgson and Kainth (2008), and Qin, Yang, Che and Feng (2026, Propositions 1 and 2) construct the power-law tails that match the boundary price and slope, whose exponents are $\bar{\gamma}$ and $\bar{\eta}$ and are unique within the power-law family; their Section 5.5 gives a necessary and sufficient condition for a shifted power-law right tail that also matches the boundary density, a condition of the same kind as, though different from, the check of setting (i). Figlewski (2010) completes a recovered risk-neutral density with generalised extreme value tails attached inside the quoted range, matching the tail probability at one quantile and the density there and at a second quantile, and Bollinger, Melick and Thomas (2023) require such tails also to price the options at the outermost strikes and show with scaled Weibull tails that this is feasible; the check of setting (i) asks the corresponding question for the classes H_L and H_U, where compatibility with the boundary price and slope requires $\gamma_i \le \bar{\gamma}$. Taleb et al. (2023, Section IV) derive from convexity at the strike where a Pareto tail is attached a restriction on its index, and Bollerslev, Todorov and Xu (2015, Section 3) use an estimator of tail decay based on log-log slopes of out-of-the-money option prices. Sharp bounds on an option price given the prices of other options, by convex optimisation and without a tail hypothesis, go back to Bertsimas and Popescu (2002). The reduction of a shape-constrained tail, with boundary values fixed by the observed part, to a moment problem over a mixing measure is that of Lam and Mottet (2017) and Mottet and Lam (2017) for convex and higher-order monotone tail densities, and the two-point extreme points are classical (Kemperman, 1968; Winkler, 1988, as quoted by Pinelis, 2016; Popescu, 2005). The breakdown point is the identification breakdown point of Horowitz and Manski (1995), as generalised by Stoye (2005, 2010) and set out by Masten and Poirier (2020), defined for the sign of skewness as Kline and Santos (2013) define it for the sign of a parameter. My own parts are the proof that $\bar{\gamma}$ and $\bar{\eta}$ are the largest exponents admissible in the whole classes H_L and H_U, with uniqueness in those classes at the maximum; the generating family of H_L and H_U with the boundary price and slope fixed, the resulting identified set and its consequences for R4; the check of setting (i); and the breakdown analysis of risk-neutral skewness. I have not found these in this form in the sources I consulted; the proofs are in the notes.[7]

2.4 The diffusive null of the hedge cost

As Jurek (2014) notes for the construction of Burnside et al. (2011), a position protected by a single put whose delta is not hedged earns only a fraction $1 + \Delta_P$ of the diffusive premium, where $\Delta_P < 0$ is the put's delta, so protection costs part of the carry premium even when no crash risk is priced. For a forward G and the put's strike K, write $d_{\pm}(G) = [\ln(G/K) \pm s^2/2]/s$.

[7] The boundary slope is the risk-neutral probability below $K_{\min}$, which finitely many quotes only bound (Zhuang, 2026); here it is the calibrated smile's, and the 10Δ prices are extrapolated from 25Δ quotes.

R6 (diffusive null of the hedge cost). *Suppose that under $\mathbb{Q}$ the terminal rate S_τ is lognormal with mean F and log-variance s^2, and that under $\mathbb{P}$ it is lognormal with the same log-variance and mean $Fe^{\lambda\tau}$, $\lambda \neq 0$, with no jumps and no volatility premium. A unit long forward has $\mathbb{P}$-expected payoff $\mu_U = F(e^{\lambda\tau} - 1)$; adding a put struck at K, bought at its $\mathbb{Q}$-value with the premium financed to τ, gives expected payoff μ_H. Then $\theta_0 = 1 - \mu_H/\mu_U = \Phi(-d_+(\xi))$ for some ξ between F and $Fe^{\lambda\tau}$, and*

$$\left|\theta_0 - \Phi(-d_1)\right| \leq \frac{1}{\sqrt{2\pi}} \frac{|\lambda|\sqrt{\tau}}{\sigma}, \qquad d_1 = d_+(F).$$

Jurek (2014, Section 2.2 of the August 2013 version) states this leading-order form, and Farhi et al. (2015, Section 5.1) obtain it as a small-maturity limit for their hedge. Jurek's own hedged trades also hedge the delta of the option overlay, so that they earn about the whole diffusive premium, and he reads his hedged-to-unhedged comparison as an upper bound on the crash share. The mean-value form and the error bound are my own short derivation. The notes treat a long forward hedged with a put; a short forward hedged with a call follows by the same argument, with the call's forward delta in place of the put's. With no crash premium at all, a put of forward delta $-\Phi(-d_1)$ surrenders approximately that fraction of the carry premium: about 0.25 for a 25Δ put and 0.10 for a 10Δ put. This is the null θ_0 against which the realised ratio θ_{UB} of Section 4 is read.

2.5 What the design measures

By R7(i), the carry premium is an entropy difference less the basis, and crash risk enters it through higher cumulants. By R1, R2 and R4, option prices on S reveal the risk-neutral law of $\Delta s = m^* - m$ at the quoted maturity, only partially beyond the 10Δ strikes, but neither the physical law nor the separate cumulants of m and m^*; they constrain, but do not identify without a parametric model of the discount factors, how $L_t(M) - L_t(M^*)$ divides across cumulant orders; Jurek and Xu (2014) impose such a model on the cross-section of G10 option prices. The design accordingly measures the price of the carry position's left-tail exposure, through the premia of options that pay when the legs lose, and does not measure the crash share of discount-factor entropy. A premium that remains after hedging can reflect diffusive risk (R6); options that are cheap as systemic insurance, which is how Caballero and Doyle (2012) explain why option-hedged carry retains high returns that systemic risk does not account for; or the pricing of a peso state (Burnside et al., 2011). The design does not separate these.

3 Data

3.1 Instruments and contributors

I obtained the option and money-market data from LSEG Workspace under a student licence provided by my university (Section 3.6). The main retrieval, of 23 September 2026, requested daily history from 1 January 1995. For each of the nine currencies (EUR, GBP, AUD, NZD, JPY, CHF, CAD, NOK and SEK, all against USD) the panel contains spot and one-month forward points (composite, bid and ask); at-the-money volatility, 25Δ and 10Δ risk reversals and 25Δ and 10Δ butterflies at one and three months, from the composite and from the Fenics contributor (bid, ask and mid); and one-month deposit rates, with OIS where an instrument exists (none exists for EUR or SEK). For USD it contains one-month Fed Funds OIS, SOFR OIS and a deposit rate. Three-month forward points and rates were retrieved on 24 September 2026 for the three-month variant, and continuation series of the first two VIX futures complete the panel. One-month 5Δ risk reversals and butterflies, which only the Fenics contributor quotes (two-sided from June 2022), were retrieved on 27 September 2026 for an out-of-sample test planned before the retrieval (Section 5.2). Named-broker contributors are short, sparse or mid-only; one is used

only as a check. WM/Reuters fixings, the S&P 500 index and the Cboe VIX index are not licensed. Mid quotes are the average of bid and ask.

The provider does not document the time of day of daily history. Matching daily values against intraday bars in winter and summer 2026 places the composite value at about 21:18 UTC throughout the year and the Fenics value at about 17:16 London time. Option payoffs use end-of-day spot on the expiry date, because the 10:00 New York cut is not available; a robustness variant uses the previous day's spot.

3.2 Samples and the month-end rule

Sampling is at New York month-ends: the last business day of the calendar month under the Federal Reserve holiday calendar (Federal Reserve Board, K.8), with a Saturday holiday not moved to the Friday. A quote is observed if it is two-sided on the month-end and substituted if the last two-sided quote lies within the five preceding business days. Otherwise it is missing and the currency is excluded that month; a month is used only if at least twice the number of legs remain. In the one-month primary sample, 24 of the 11,520 quote fields (eight per currency-month) were substituted, in 23 currency-months, and no currency-month was excluded. The primary sample uses composite quotes. It starts at the first month-end at which all 45 composite one-month smile series (at-the-money, 25Δ and 10Δ risk reversals and butterflies for nine currencies) and the nine forward-point series are available under this rule, and runs from May 2013 to August 2026: 160 month-ends, of which 104 fall in the zero-rate regime (May 2013 to December 2021) and 56 in the hiking regime (from January 2022). The return window of the last month-end ends after the last spot observation and is excluded, so the return sample has 159 month-ends, May 2013 to July 2026, with returns realised from June 2013 to August 2026.

The extended sample uses Fenics quotes before the primary start, with two long and two short legs, from January 2007 to April 2013; 72 of these month-ends have the required four currencies with the calibration set (at-the-money, 25Δ risk reversal and butterfly) and forward points. Fenics quotes before 2010 update infrequently, butterflies in particular, so results from this sample are secondary evidence. The design also defines a long at-the-money sample, at-the-money-hedged carry as in Burnside et al. (2011) from the start of the at-the-money and forward series. It uses composite spot, forward points and at-the-money volatility only, with three long and three short legs when at least six currencies are available and two and two when four or five are, and the month is excluded otherwise. Its first month-end, January 1995, is the first of the provider's one-month at-the-money series: a request on 1 October 2026 for history from 1970 returned no at-the-money quote before 6 January 1995, although spot quotes go back to 1971 and forward points to 1982. The sample has 379 month-ends with realised returns to July 2026, the first 36 with two legs per side. I fixed its rules after the E2, E3 and E5 results had been recorded: its 159 primary-window months reproduce the at-the-money-hedged returns of Table 7 exactly, and the Fenics at-the-money series of the extended sample was also known, but no return before May 2013 had been computed under these rules (Section 4.7).

3.3 Quote conventions and audit outcomes

Quote conventions follow Reiswich and Wystup (2012). From the conventions of Clark (2011) that they report, I take spot delta, which pairs of these currencies use up to and including one year, and premium-adjusted delta when the premium is paid in the base currency, pips delta otherwise. Their table gives USDJPY, USDCHF and USDCAD as premium-adjusted and EURUSD, GBPUSD and AUDUSD as pips delta; the conventions for USDNOK and USDSEK (premium-adjusted) and NZDUSD (pips) follow from the premium-currency hierarchy they report, in which the dollar ranks first. At-the-money is the delta-neutral straddle, with strike $F\exp(\pm\sigma^2\tau/2)$ for pips and premium-adjusted delta respectively (their Table 3), and the risk reversal is call minus put volatility on the quoted pair. For dollar-base pairs, spot and forward

are inverted so that X_t^j is the USD price of one unit of currency j; the outright forward is spot plus forward points times the pip factor.

The data audit checked coverage, two-sidedness, staleness, the underlying pair, quote semantics, conventions, the splice between contributors and plausibility (implausible quotes are flagged, not removed). Provider metadata give every volatility instrument, including those for NOK and SEK, as the currency against USD, and a realised-volatility test agrees, so no triangulation through EURUSD is needed. The forward-point scaling factors equal the configured pip factors. Risk reversals are call minus put on the quoted pair, and the 10Δ quotes follow the 25Δ convention. Instrument metadata name only the quote type and the delta, and LSEG's documentation of its own surfaces does not define the contributed quotes (LSEG Developer Portal), so delta and premium conventions follow market convention as above. Fenics and composite skew prices agree leg by leg, with correlations of 0.995 before the primary sample and 0.997 within it. A second retrieval of every instrument over the same dates, on 26 September 2026, returned byte-identical files for all 243 instruments with data, so no quote was revised between my retrievals; the check cannot detect revisions made before the first. The same retrieval's metadata describe the three-month USD OIS series of the three-month variant as the Fed Funds OIS rate.

3.4 The butterfly reading

The quoted 25Δ butterfly BF can be read in two ways. Under the market-strangle reading, a strangle struck at the 25Δ strikes of the flat volatility $\sigma_{ATM} + BF$ has the same premium under the smile as at that flat volatility. Under the smile-strangle reading, BF is the average of the smile's volatilities at its own 25Δ strikes less σ_{ATM}. A three-parameter smile fits either reading exactly, so the quotes cannot decide between them. The design sets the reading by provider documentation first, then by the market-strangle convention (Reiswich and Wystup, 2012), which the design names as the interdealer convention, with the fit to the held-out 10Δ quotes as supporting evidence only. Provider documentation is silent, so the market-strangle reading is primary. The supporting diagnostic points the other way: smiles calibrated under the smile-strangle reading predict the held-out 10Δ butterflies better in most month-ends and in every currency, with small absolute differences. This is recorded as evidence against the primary reading, and E4 (smile-strangle reading) is reported alongside the primary results.

3.5 VIX futures

Individual VIX futures contracts are taken from the Cboe Futures Exchange historical settlement files, one per contract, for contracts from January 2007 to December 2026. Before 26 March 2007 the archive quotes contracts at ten times the index level; the change of scale is detected in the files and removed. After this, Cboe second-month settlements equal the LSEG second-month continuation at every month-end (223 of 223). The Cboe files are stored privately because their redistribution terms are not established.

3.6 Licence

I obtained the LSEG data under an LSEG Workspace student licence provided by my university. The licence covers only me: it permits individual study and research, and it does not permit redistribution of the data or placing them in a public repository. Neither this paper nor the repository therefore contains raw LSEG data. They publish code, methods and aggregate results: means, standard errors, test statistics, intervals and coverage facts such as sample windows and counts. My university, the licence holder, has since confirmed that manipulated or transformed data may be published and that the raw data may not. The pooled results reported here are transformed data in that sense, and so are the month-level series at the level of the carry portfolio

that the repository publishes. From them the main portfolio-level results can be recomputed: E1 under the market reading with 10Δ hedges, E2 with φ, the secondary moment predictors, E3 and its split by regime, E5, the hedge-cost ratio and its confidence sets, the long at-the-money sample apart from two details, the comparison of the regime changes in the skew price and the spread, the base and at-the-money rows of the robustness grid and the returns of its 25Δ row; Appendix B lists what needs the quotes. Each published value averages at least four currencies whose identities are not published, and each month publishes fewer values than the quotes it depends on. Series that would have let an adversary who knew the legs and the exact forwards narrow individual risk-reversal quotes were withheld, and the repository's publication policy records the test applied. Quote values, per-currency series and calibrated smile parameters are held privately; the parameters are not a substantive transformation, because a calibrated smile reproduces the quotes it was fitted to, so they would let a reader recover the raw quotes. The figures plot only these published portfolio-level series, aggregate results and a synthetic model. Every table and figure of market data is computed from LSEG Workspace data, and the calculations are mine. Anyone else who wants to rerun the acquisition, and hence the estimation, needs their own LSEG Workspace licence.

4 Method

4.1 Smile calibration

For each currency, month-end and butterfly reading, a SABR smile with $\beta = 1$ (Hagan, Kumar, Lesniewski and Woodward, 2002) is calibrated to the at-the-money volatility, the 25Δ risk reversal and the 25Δ butterfly. Under the primary reading the three equations are: the smile's volatility at the delta-neutral-straddle strike equals σ_{ATM}; the difference of the smile's volatilities at its own 25Δ call and put strikes equals the risk reversal; and the market strangle has the same premium under the smile as under the flat volatility $\sigma_{ATM} + BF$ (Reiswich and Wystup, 2012). The system is solved by Levenberg–Marquardt in $(\ln\alpha, \operatorname{atanh}\rho, \ln\nu)$, starting from the previous month's solution with three fixed starting points as fall-backs. A calibration has converged when the largest absolute residual is below 10^{-8} volatility units. The 10Δ quotes are held out as a fit test. Strikes are found from deltas in each pair's own convention.

R5 (delta-to-strike inversion). *(a) For pips spot delta $\Delta = \omega D_b \Phi(\omega d_+)$ of a call ($\omega = 1$) or put ($\omega = -1$) at a flat volatility, with D_b the base-currency discount factor, $K = F\exp\big(\sigma^2\tau/2 - \omega\sigma\sqrt{\tau}\,\Phi^{-1}(|\Delta|/D_b)\big)$ for $0 < |\Delta| < D_b$ with Δ of the sign of ω. (b) The premium-adjusted call delta $h(K) = D\,(K/F)\,\Phi(d_-(K))$, with D the base-currency discount factor for spot delta and $D = 1$ for forward delta, tends to zero at both ends, is strictly increasing up to a unique maximiser K^* and strictly decreasing after it; each $\Delta \in (0, h(K^*))$ is attained at exactly two strikes, and the convention takes the one in $[K^*, \infty)$. (c) For a continuously differentiable smile $\sigma(K)$, if $|\partial\sigma/\partial k|\sqrt{\tau}\,|d_-(K, \sigma(K))| < 1$ on an interval, with $k = \ln(K/F)$, the pips delta $\omega D\Phi(\omega d_+(K, \sigma(K)))$ is strictly decreasing there, so each delta attained on the interval is attained at exactly one strike.*

Reiswich and Wystup (2012) give the delta definitions, the closed-form strike for pips deltas, the non-monotonicity of the premium-adjusted call delta, the right-branch convention and the first-order condition for its maximum. Jäckel (2020, Section 2, eq. (20)) gives the single maximum in (b) through the inverse Mills ratio, the argument the notes write out.[8] Reiswich (2010, Section 3.8) shows that a smile delta can fail to be monotone, and Healy (2025, Section 5.3) gives SABR smiles with $\beta = 1$ whose pips call delta is not monotone and never reaches 25Δ. The condition in (c) is the two-sided form of the inequality of Fukasawa (2012, Lemma 2.1), which in this notation

[8]That the inverse Mills ratio is decreasing follows from the log-concavity of Φ (Bagnoli and Bergstrom, 2005) and is classical (Sampford, 1953).

reads $1 + \sqrt{\tau}\, d_- \, \partial\sigma/\partial k > 0$ and from which his Proposition 2.1 obtains the monotonicity of d_+ by the derivative identity the notes use; Mingone (2023, Lemma 4.1) shows that each delta has a unique strike if and only if d_+ is strictly decreasing along the smile and maps onto the real line. Fukasawa derives the inequality from the absence of arbitrage; here it is a hypothesis checked numerically on each calibrated smile, much as Martini and Mingone (2022) impose the same conditions on parametric smiles that need not be free of arbitrage. Nothing in (c) is mine beyond this check. Across the calibrated one-month smiles of the primary sample, the condition in (c) holds over ±4 at-the-money standard deviations at every month-end, and premium-adjusted deltas are monotone on the conventional branch over the same range at every month-end.

R8 (local well-posedness of the calibration). *For $\beta = 1$ the Hagan implied volatility is infinitely differentiable in (K, α, ρ, ν) on $\{K > 0\} \times (0, \infty) \times (-1, 1) \times [0, \infty)$. Let $\Psi(\theta, q) = 0$ be the three calibration equations in $\theta = (\alpha, \rho, \nu)$ for quotes $q = (\sigma_{ATM}, RR, BF)$, with a solution (θ^*, q^*). Suppose that at the solution (i) the at-the-money strike equation has a non-zero strike derivative; (ii) the smile delta has a non-zero strike derivative at the 25Δ strikes; (iii) for premium-adjusted delta, the 25Δ call strike at the flat volatility $\sigma_{ATM} + BF$ lies strictly right of the maximiser K^*; and (iv) $D_\theta \Psi$ is non-singular. Then there are neighbourhoods $U \ni q^*$ and $W \ni \theta^*$ and a unique C^1 map $\theta : U \to W$ with $\Psi(\theta(q), q) = 0$.*

The proof applies the implicit function theorem first to the strikes and then to the system, following the argument Reiswich (2010, Theorems 2 and 3) uses for the volatility-strike function and for the market-strangle calibration of a simpler smile. The three equations are the conditions of Reiswich and Wystup (2012), to which Reiswich (2010, Section 3.2) calibrates SABR, and Henrard (2011) uses the implicit function theorem in the same way for any exactly determined calibration.[9] Clark (2011, Sections 3.7 and 3.10) calibrates SABR with $\beta = 1$ to these three conditions by minimising an error function numerically, notes that the quotes prescribe no unique smile, and states no result on the existence or uniqueness of the parameters. Hagan et al. (2002, Section 3.1) observe informally that, with β fixed, α, ρ and ν play widely separated roles, so that fitted values are stable, while β and ρ are hard to distinguish, and Reiswich (2010, Section 3.2.2) finds the least-squares calibration of SABR with β free to the same three conditions ill-posed, with flat directions of the objective; R8 fixes $\beta = 1$ and requires condition (iv) at the solution instead. The uniqueness in R8 is within the SABR family: exact interpolations of the same quotes in other families give different 10Δ volatilities (Healy, 2025, Section 3). What I claim as my own is the joint smoothness statement for $\beta = 1$, including $\nu = 0$, and the local well-posedness statement for SABR under conditions (i) to (iv), written for FX delta conventions; I have not found these in this form in the sources I consulted. Because the result is local, the solver restarts from several points; condition (iv) is checked at every calibration through the solver's Jacobian, whose condition numbers confirm non-singularity everywhere. Under the smile-strangle reading the result holds without (iii).

Every smile is checked for static arbitrage (R2) on log-moneyness grids of ±4 and ±10 at-the-money standard deviations, by discrete convexity of call prices and by the density condition of Gatheral and Jacquier (2014). Non-convergence and arbitrage flags are counted, not dropped. In the primary sample every month-end smile converged under both readings, with residuals below 10^{-13}, and every smile passed both checks.

[9] Le Floc'h and Kennedy (2014, Section 4, eqs. (30)–(31)) invert the leading-order expansion of the Hagan formula (Hagan et al., 2002, eq. (3.1a)) in closed form, recovering (α, ρ, ν) from the at-the-money level, skew and curvature, which is an approximate, global counterpart of the local and exact statement here. In the smoothness lemma, the value of the ratio $z/x(z)$ at $z = 0$ is that of the at-the-money formula of Hagan et al. (2002, eq. (2.18)), Choi and Wu (2021, eq. (5)) give its Taylor expansion there, and the integral representation is Hadamard's lemma (Milnor, 1963).

4.2 Portfolios and protective options

At each month-end t the available currencies are ranked by the forward discount $fd_t^j = s_t^j - f_t^j = \log(X_t^j/F_t^j)$, where F_t^j is the one-month outright forward. Lustig, Roussanov and Verdelhan (2011) sort currencies into six portfolios on the forward discount at each month-end and take the highest minus the lowest; with nine currencies the analogue is long the top $n = 3$ and short the bottom $n = 3$, each with $1/n$ USD of forward notional ($n = 2$ in the extended sample). With arithmetic excess returns $rx_{t+1}^j = X_{t+1}^j/F_t^j - 1$ per USD of forward notional,

$$HML_{t+1}^U = \frac{1}{n}\sum_{\text{long}} rx_{t+1}^j - \frac{1}{n}\sum_{\text{short}} rx_{t+1}^j, \qquad FD_t = \frac{1}{n}\Big(\sum_{\text{long}} fd_t^j - \sum_{\text{short}} fd_t^j\Big) \geq 0. \tag{2}$$

Returns are evaluated at the spot quote on the option expiry date, the spot rate for value on the forward's delivery date. Dates follow the New York business calendar, and volatility time is trade to expiry in days over 365.

Each leg buys a one-month option that pays when the leg loses: a put on the currency for a long leg, which is a USD call for dollar-base pairs, and a call on the currency for a short leg. The hedge follows the construction of Burnside et al. (2011, Section 1.2): option notional matches the forward notional, and premia are compounded forward, here to the delivery date. Premia are therefore in USD per USD of forward notional at delivery, which for both quotation types equals the undiscounted premium on the quoted pair divided by the pair forward. The primary strike is the 10Δ strike on the calibrated smile in the pair's own delta convention (R5); 25Δ and at-the-money hedges are comparisons. Pricing is Garman and Kohlhagen (1983) in forward form. The quote-currency discount factor comes from Fed Funds OIS for USD and one-month deposit rates otherwise, and the base-currency factor is implied by the forward, $D_b = FD_q/S$, so no separate covered-interest-parity assumption is made. With Π_{t+1} the option payoff per USD of notional and V_t^{smile} its premium at the smile volatility,

$$HML_{t+1}^H = HML_{t+1}^U + \frac{1}{n}\sum_{\text{legs}}\big(\Pi_{t+1} - V_t^{\text{smile}}\big). \tag{3}$$

Estimands use mid prices. An implementable version trades forwards at bid and ask and buys options at the smile volatility plus $k \in \{1, 1.5, 2\}$ times the at-the-money half-spread, converted to premium by vega, because package quotes do not give a single-option ask.

4.3 Estimands

The primary estimand, E1, is measured ex ante. For each leg, V_t^{smile} is the premium of the 10Δ protective option at the calibrated smile volatility and V_t^{flat} the premium at the same strike with the at-the-money volatility. Then

$$C_t^{\text{skew}} = \frac{1}{n}\sum_{\text{legs}}\big(V_t^{\text{smile}} - V_t^{\text{flat}}\big), \qquad \varphi_t = C_t^{\text{skew}}/FD_t, \tag{4}$$

the skew price of protection per unit of the forward discount of equation (2). E1 reports the means of φ_t in the zero-rate and hiking regimes and their difference. The 25Δ version is a comparison. The at-the-money comparison named in the design is identically zero, because the calibrated smile passes through the at-the-money volatility, and is not reported.

E2 asks whether the skew price predicts carry returns, through the regression $HML_{t+1}^U = a + b\, z_t + e_{t+1}$ with $z_t = \varphi_t$ (primary) or the oriented 10Δ risk reversal: the quoted volatility of the protective option minus that of the opposite option, averaged over legs. The out-of-sample test uses an expanding window, with the first forecast, made at the end of December 2016, for January 2017 and training that starts with the extended sample. Option-implied variance and skewness are secondary predictors, reported as R4 intervals.

E3 is measured ex post. The realised excess payoff of each option splits into three terms, all carried to delivery:

$$\Pi - V^{\text{smile}} = \underbrace{\left[\Pi - V^{\text{GK}}(\hat{\sigma}^P)\right]}_{\text{(i) payoff}} \underbrace{-\left[V^{\text{flat}} - V^{\text{GK}}(\hat{\sigma}^P)\right]}_{\text{(ii) volatility level}} \underbrace{-\left[V^{\text{smile}} - V^{\text{flat}}\right]}_{\text{(iii) skew}}, \tag{5}$$

where $V^{\text{GK}}(\hat{\sigma}^P)$ is the Garman–Kohlhagen premium at the same strike with $\hat{\sigma}^P$, the annualised realised volatility of daily log spot changes over the previous 21 business days. Term (i) carries the delta and diffusive part; term (iii) is known at t. Averaged over legs with weight $1/n$, the three terms add up to $HML^H - HML^U$ of equation (3). The hedge-cost ratio is

$$\theta_{\text{UB}} = 1 - \hat{\mu}_H / \hat{\mu}_U, \tag{6}$$

with $\hat{\mu}_H$ and $\hat{\mu}_U$ the sample means of HML^H and HML^U; it can take any real value. Jurek (2014) reads such a ratio, from trades whose option delta is itself hedged, as an upper bound on the crash share; an unlevered hedge also gives up part of the diffusive premium, so θ_{UB} is compared with θ_0 of R6 and is not described as a crash share. Here θ_0 is the equal-weight average over legs of the absolute forward delta of the hedging options; for the portfolio, R6 weights each leg by its expected forward payoff, so the equal weights are an approximation.

E4 recomputes E1 and the skew term of E3 under the smile-strangle reading of the butterfly quote.

E5 measures exposure to systemic risk: HML^U and HML^H are regressed on two factors, both measured over each carry return window from the month-end to the one-month option expiry. The VIX roll-down is minus the percentage change of the second-month VIX future's settlement price, holding the same contract. It is a month-end analogue of the strategy of Caballero and Doyle (2012), who short the VIX future whose expiry matches the one-month forward's maturity and hold it to expiry. The global FX-volatility level follows Menkhoff, Sarno, Schmeling and Schrimpf (2012, eq. (4)): the average over the window's days of the cross-sectional mean absolute daily log spot change of the nine currencies. Its innovation is the residual of an AR(1) fitted to that series. E5 reports the exposure-adjusted ratio $1 - \alpha_H/\alpha_U$.

The design also names a descriptive account of smiles, φ_t and returns around 5 August 2024; Section 5.4 reports it from averages over the portfolio's legs and the nine currencies.

4.4 Inference

Standard errors are those of Newey and West (1987) with a Bartlett kernel, and the bandwidth follows the automatic procedure of Newey and West (1994) without prewhitening: a pilot truncation of $\lfloor 4(T/100)^{2/9} \rfloor$, then a bandwidth of $\lfloor \hat{\gamma} T^{1/3} \rfloor$. The E1 difference is the coefficient on a regime dummy. Its bootstrap interval uses the stationary bootstrap of Politis and Romano (1994), resampling each regime separately with its own expected block length chosen by the rule of Politis and White (2004) as corrected by Patton, Politis and White (2009), with 9,999 draws and percentile intervals. The block-length routine was written with the arch package (Sheppard, 2026) as the reference implementation and follows its steps, including its caps (arch states that some of its tuning parameters are taken from Andrew Patton's MATLAB program); it departs from arch in two stated conventions and one degenerate case. The bandwidth code was likewise written with the R package sandwich (Zeileis and Lumley, 2026) as the reference for the constants and for the zero weight on the intercept.

For the predictive regressions, a residual bootstrap under the null of no predictability, in the manner of Nelson and Kim (1993) and with 9,999 draws, draws the pairs $(\hat{u}_k, \hat{v}_{k+1})$ independently, where $\hat{u}$ are residuals of the unrestricted regression and $\hat{v}$ are AR(1) residuals of the predictor. The slope is corrected by the bootstrap bias estimate, which is checked against the first-order approximation $-(\sigma_{uv}/\sigma_v^2)(1+3\rho)/T$ of Stambaugh (1999, eq. (18)). The out-of-sample test is the one-sided MSPE-adjusted statistic of Clark and West (2007) against the historical mean,

with the least-squares standard error they recommend for one-step forecasts, since the design names none; the Newey–West version is reported alongside.

The 95% set for θ_{UB} is obtained, as the design specifies, by test inversion: $\{\vartheta : |\hat{\mu}_H - (1 - \vartheta)\hat{\mu}_U| \leq 1.96\, \widehat{se}_{HAC}(\vartheta)\}$, with $\widehat{se}_{HAC}(\vartheta)$ the Newey–West standard error of $\hat{\mu}_H - (1 - \vartheta)\hat{\mu}_U$. This is the construction of Fieller (1954, Sections 2–4) for a ratio of means, with a Newey–West variance in place of his independent variance estimate: the set is a bounded interval only when the denominator $\hat{\mu}_U$ is significantly different from zero, and is otherwise the complement of an interval or the whole line. Unbounded sets are reported as unbounded. Because the automatic bandwidth does not depend on the scale of the series, the statistic tends to $|\hat{\mu}_U|/\widehat{se}_{HAC}(\hat{\mu}_U)$ as $|\vartheta|$ grows, so the set is unbounded on both sides when that t-statistic is below 1.96 in absolute value and bounded when it is above, whatever finite grid is scanned. The design names delta-method intervals as secondary; Section 5.3 reports them to show how they mislead here. Because E1 is the single primary estimand, no multiple-testing adjustment is made.

R9 (inference; cited). *(a) If (X_t) is strictly stationary with $\mathbb{E}|X_t|^{2+\delta} < \infty$ for some $\delta > 0$, strong-mixing coefficients with $\sum_n \alpha(n)^{\delta/(2+\delta)} < \infty$ and a positive long-run variance, its sample mean is asymptotically normal (Ibragimov, 1962, Theorem 1.7). (b) The stationary bootstrap is valid under the moment and mixing conditions of Theorem 2 of the technical-report version of Politis and Romano (1994); Gonçalves and de Jong (2003) weaken them. (c) Confidence sets for a ratio that have finite expected diameter have zero worst-case coverage (Gleser and Hwang, 1987), and so do sets that are bounded with probability one (Dufour, 1997), hence test inversion.*

These conditions are assumptions, not checked properties. The persistence of φ and the heavy right tail of its zero-rate distribution bear directly on them, and each regime mean averages a short, dependent stretch of calendar time; the post hoc analyses of Section 5.1 address this only in part.

4.5 Interpretation rules

Section 9 of the design fixes how results are read. A regime difference in E1 is claimed only if its 95% interval excludes zero, and its size is read relative to FD_t. The design reports a Newey–West and a stationary-bootstrap interval without saying which governs; I treat a difference as significant only when both exclude zero. This reading is mine, not the design's; I adopted it when interpreting the robustness grid, after the primary results, and the research log records it. It does not affect the primary E1 result, where both intervals include zero, but it decides the dollar-carry variant and the three three-month phases, in which only the bootstrap interval excludes zero (Section 6). The statement that crash-insurance pricing explains the regime shift requires all three of: the E1 difference, $b > 0$ in E2 after the Stambaugh correction, and an out-of-sample Clark–West rejection at 5%; otherwise the result is reported as partial or no support. θ_{UB} is not described as a crash share, and a skew term whose interval includes zero is reported as no evidence of priced crash risk beyond the diffusive and volatility-level terms. Null results are reported with the same prominence as positive ones.

4.6 Validation

Pricing and delta conventions are checked against analytic identities (put–call parity, the delta-neutral straddle and delta-to-strike round trips), and smile calibration by synthetic recovery under both readings and both delta types. QuantLib 1.43 (QuantLib Team, 2026), an independent open-source implementation, agrees with the project's premia, deltas in three conventions, strike inversion, delta-neutral strikes and Hagan volatilities to between 10^{-9} and 10^{-13} on the test grid.[10] Before the E1 results were recorded, a separate review of the estimand and inference code

[10] Calibration uses the Levenberg–Marquardt routine of SciPy 1.16.3 (MINPACK), and the R10 identified sets are linear programmes solved with HiGHS (Huangfu and Hall, 2018) through highspy 1.15.1.

rebuilt every leg in QuantLib, which agreed to machine precision, and found that the standard errors matched sandwich exactly. The base variant of the robustness grid reproduces the E1 regime difference and the means of HML^U, HML^H and the skew term to machine precision. The stationary bootstrap and its block-length rule are checked numerically against the arch package (Sheppard, 2026, release 8.0.0). The two differ by design in two choices, the lag at which the window of insignificant autocorrelations starts and the normalisation of the autocorrelations; with these switched to arch's, the block lengths are identical, and the two resamplers agree in distribution. Recomputing every E1 interval entirely with arch, its own block lengths included, moves no bound by more than 0.0094 for φ or 0.031 bp for C^{skew} and FD, and changes no conclusion.

The moment code reproduces, from a flat smile, the lognormal moments implied by the dynamics of Garman and Kohlhagen (1983) when nothing is truncated, a test in the suite, and is validated against Merton (1976) and Heston (1993) models with one-month FX-typical parameters, priced in QuantLib (Merton through its Bates engine with constant variance, which agrees with Merton's eq. (19) to within $10^{-14}F$, and Heston through its analytic engine, which agrees with direct Fourier prices to within $3 \times 10^{-16}F$),[11] with exact moments from the cumulant generating functions (for Heston through the second of the two forms of the characteristic function compared by Albrecher, Mayer, Schoutens and Tistaert, 2007, Section 3). In all eight model and pair combinations, spanning over the model smile reproduces the model moments and the R4 intervals contain the true moments, which the test suite asserts, and the R10 identified sets at the model's own tail exponents contain them (Table 1). These analyses are synthetic and involve no LSEG data.

As the design specifies, HML^U is compared in sign and magnitude with the public developed-currency portfolios of Lustig, Roussanov and Verdelhan (2011), from the file on Verdelhan's data page (Verdelhan, 2021), which ends in May 2021. Over the 96 overlapping months from June 2013, with each of my returns paired with their return for the following month-end, the means are 7.8 bp per month (Newey–West s.e. 16.1) here and 7.2 bp (s.e. 26.5) there, the correlation is 0.51, the signs agree in 77% of months and the slope of HML^U on their series is 0.41 (s.e. 0.12). The two portfolios differ in currencies, in the number of portfolios and in timing, so agreement in sign and magnitude is what the check can show.

Four further checks, named in the design or, for the named-broker contributor, in the data plan, were computed after the main results (Section 4.7). Discounting the dollar with SOFR OIS instead of Fed Funds OIS from June 2018, when the SOFR series starts under the month-end rule, changes the base-currency discount factor by at most 0.013%, φ by at most 1.1×10^{-4} and the skew price by at most 0.002 bp in any month, because the dollar rate enters E1 only through that factor; recalibrating every affected smile, the E1 difference is -0.3548 under both rates and no conclusion changes. Over 96 EUR month-ends from July 2018, the named-broker contributor's at-the-money volatility is on average 0.02 volatility points below the composite's (s.d. 0.09) and its 25Δ risk reversal and butterfly differ by less than 0.005 points on average. Smiles fitted to its quotes change the EUR leg's 10Δ skew price by 0.07 bp per month on a mean of 2.7 bp, and the portfolio's skew price by 0.03 bp on 11.5 bp; the broker quotes no 10Δ strikes, so this compares two extrapolations from 25Δ quotes. In the moment code, the adaptive tail quadrature agrees with the closed-form R4 bounds to 1.6×10^{-15} relative in all 1,440 currency-months, and Simpson's rule for the middle part converged in all of them. Finally, a C++ kernel reimplements the premia, deltas, strike inversion, SABR volatilities and the middle part of the moment integrals, operation by operation as in the Python code, which remains the code of record. Parity tests on synthetic inputs bound the differences at 10^{-13} relative. For

[11] The Fourier prices and tail probabilities use the formulas of Lee (2004b, Theorem 5.1) on the contour of Lord and Kahl (2007, Section 3.2); for Heston the moment strip uses the explosion time of Keller-Ressel (2011, Section 6.1), and derivatives of the cumulant generating function use trapezoidal Cauchy integrals (Bornemann, 2011). In the check that spanning over the model smile reproduces the model moments, the tails beyond the outermost strikes of the model smile are bounded with the price bounds of Lee (2004a, Theorem 2.1).

Table 1: Model validation of the tail hypotheses (synthetic models)

Model	Pair	Skew	θ_{true}	(i)	Check	Sign	θ^*
Merton, down jumps	EURUSD	-1.578	0.85, 0.58	fails	fails	yes	0.21
	USDJPY	1.471	0.84, 0.58	fails	fails	yes	0.26
Merton, symmetric jumps	EURUSD	-0.000	0.61, 0.61	fails	fails	no	1.00
	USDJPY	-0.077	0.61, 0.61	fails	fails	no	0.98
Heston, $\rho = -0.3$	EURUSD	-0.638	0.97, 0.96	fails	holds	yes	0.49
	USDJPY	0.592	0.97, 0.96	fails	holds	yes	0.54
Heston, $\rho = +0.3$	EURUSD	0.477	0.92, 0.95	holds	holds	yes	0.56
	USDJPY	-0.507	0.92, 0.94	fails	holds	yes	0.52

One-month models: Merton with diffusive volatility 0.08, jump intensity 2 (down jumps: mean log jump -0.04, s.d. 0.06) or 1 (symmetric: mean 0, s.d. 0.05); Heston with initial and long-run variance 0.01, mean reversion $\kappa = 2$ and vol of variance 0.5 ($\rho = -0.3$) or 0.4 ($\rho = 0.3$). Skew: skewness of $\log(X_T/F_X)$ under the USD forward measure. θ_{true}: the model's largest valid exponent in the lower and upper tail, the infimum of the reversed hazard (lower) and hazard (upper) rate of $\ln S_T$ beyond the 10Δ strikes, as a fraction of $\bar{\gamma}$ and of $\bar{\eta} - 1$ (R10(a)). (i): whether the boundary elasticities of setting (i) satisfy R4's hypotheses in the model. Check: whether $\gamma_i \le \bar{\gamma}$ and $\eta_i \le \bar{\eta}$. Sign: whether the sign of the skewness is identified at the model's own exponents. In every row the R10 identified set at the model's own exponents contains the true variance and third central moment. θ^*: breakdown point of the sign (R10(e)); the sign at $\theta = 1$ agrees with the true sign wherever the skewness is not close to zero.

premia the bound is normwise, because an out-of-the-money premium is a difference of nearly equal terms, which amplifies the difference of some twenty units in the last place between the two implementations of Φ. On one core the kernel is about 1.2 to 2.7 times faster than the vectorised NumPy code, 3 to 9 times faster for the middle part of the moment integrals and 70 to 350 times faster for strike inversion, where the Python code loops.

4.7 Departures from the design

The research log records every departure from the design with its date, and Appendix C lists them. Three matter for reading the results. First, analyses added after the first results, among them the split of E1, the comparison of the regime changes, the identification analysis with R10 and every test of $\theta_{\text{UB}} = 0$, are post hoc and are labelled so wherever they appear. Second, the robustness grid trains the out-of-sample test of E2 from the primary start, not from the extended sample as the design fixes, because several variants cannot be built on the extended sample; that rule was fixed before any variant was estimated. Third, a check of every design item on 2 and 3 October 2026 found items that had not been computed, which I then computed under rules fixed in their scripts before they were run, and bootstraps run with 1,999 draws instead of the design's 9,999, which I reran; no conclusion changed at the 5% level.

5 Results

All results are aggregate statistics over the primary sample unless stated. Means and E3 terms are in basis points (bp) of notional per month; the units of regression coefficients are given with each table.

5.1 E1 and E4: the skew price of protection across regimes

Table 2 reports E1 and E4. Under the market-strangle reading with 10Δ hedges, the point estimate of φ falls from 0.809 to 0.454, but both intervals for the difference include zero, under both readings and at both hedge deltas, so the regime difference is not claimed. The design reads its size relative to FD_t. A supplementary (post hoc) split, Panel B, separates the two components of the ratio: there is no evidence that the skew price of protection changed (difference -0.05 bp, interval $[-2.44, 2.20]$), whereas the forward-discount spread is higher in the hiking regime

Table 2: E1 and E4: regime means and differences

Panel A: pre-registered estimand φ and comparisons						
Reading, hedge	Quantity	Zero-rate	Hiking	Difference	HAC s.e.	Bootstrap 95%
Market, 10Δ (E1)	φ	0.809 (0.185)	0.454 (0.066)	−0.355	0.204	[−0.814, 0.014]
Market, 25Δ	φ	0.613 (0.152)	0.341 (0.048)	−0.272	0.166	[−0.642, 0.027]
Smile, 10Δ (E4)	φ	0.781 (0.178)	0.437 (0.062)	−0.345	0.196	[−0.787, 0.010]
Smile, 25Δ	φ	0.595 (0.148)	0.330 (0.046)	−0.265	0.161	[−0.624, 0.024]
Panel B: supplementary (post hoc) split of the E1 ratio						
Market, 10Δ	C^{skew}	11.24 (0.82)	11.19 (0.83)	−0.05	1.30	[−2.44, 2.20]
	FD	18.61 (1.77)	27.05 (1.41)	8.44	2.30	[3.48, 13.65]

Market: market-strangle reading; smile: smile-strangle reading. Primary sample, May 2013 to August 2026: 104 zero-rate and 56 hiking month-ends. φ is dimensionless; C^{skew} and *FD* in bp per month. Newey–West standard errors in parentheses. The interval is the stationary-bootstrap 95% percentile interval for the difference (hiking minus zero-rate), 9,999 draws. Panel B was added after the first results and does not replace the pre-registered estimand.

(difference 8.44 bp, interval $[3.48, 13.65]$). The point estimate of the fall in φ therefore reflects the forward-discount spread, which was smallest in 2020–2021: excluding those years, the difference is −0.03 (post hoc, below). Figure 1 shows the two components month by month. The skew price stays near its mean apart from the month-end of March 2020, when it reached 47.5 bp, while the spread is below 10 bp at 21 month-ends between April 2020 and January 2022 and is higher throughout the hiking regime. Against the rays of constant φ in panel (b), the hiking months lie to the right of the zero-rate months at about the same height, and every month with φ above 1.5 falls in 2020 or 2021, when the spread was smallest.

Under the smile-strangle reading (E4) the difference is −0.345 (standard error 0.196, interval $[-0.787, 0.010]$); the reading moves φ by less than 4 per cent and the skew term of E3 from −11.2 to −10.8 bp per month (Table 9), so the bias from misreading the butterfly convention is small and changes no conclusion.

I added several supplementary analyses after the first E1 results, and they are reported as post hoc. They were prompted by two features of the series: φ is highly persistent and its zero-rate distribution has a heavy right tail from months with very small forward discounts; they do not replace the pre-registered estimand. The first-order autocorrelations of φ are 0.85 and 0.76 in the two regimes, leaving roughly 8 effective observations per regime.[12] With so few, the automatic Newey–West bandwidth is capped well below what the persistence would require, so the pre-registered intervals are likely too narrow. An AR(1) plug-in standard error of the difference is 0.27 ($t = -1.31$). Excluding 2020–2021, the zero-rate period with the smallest forward discounts, the difference is −0.03. The median of φ falls from 0.53 to 0.38 and the 10% trimmed mean from 0.63 to 0.41; the mean of $\log \varphi$ falls from −0.46 to −0.89, and the ratio of regime means, mean C^{skew} over mean *FD*, from 0.60 to 0.41. The Newey–West standard error of the difference rises with the lag, from 0.079 at lag 0 to 0.204 at lag 9 and 0.236 at lag 20, and the two-sided bootstrap *p*-value is 0.064. Dropping the largest 1, 3, 5 and 10 zero-rate observations gives differences of −0.32, −0.28, −0.23 and −0.14. The excluded 2020–2021 period contains 24 of the 104 zero-rate month-ends.

A further post hoc comparison asks directly whether the rise in the forward-discount spread was matched by a rise in the price of crash insurance, comparing the regime changes of the numerator and denominator of φ instead of the change in the ratio. The skew price did not change (−0.05 bp, s.e. 1.30, AR(1) plug-in s.e. 1.29), nor did the oriented 10Δ risk reversal of the portfolio (0.21 volatility points, s.e. 0.29, bootstrap interval $[-0.34, 0.67]$), while the rise in the spread survives a persistence-robust standard error ($t = 2.80$). The ratio of the two changes, $\Delta C^{\text{skew}}/\Delta FD = -0.006$, is the grouping estimator of Wald (1940) with the regime as

[12] The effective sample size of a mean under AR(1) dependence is $T(1-\rho)/(1+\rho)$, and the plug-in standard error uses the AR(1) long-run variance $\gamma_0(1+\rho)/(1-\rho)$, with γ_0 the variance of the series.

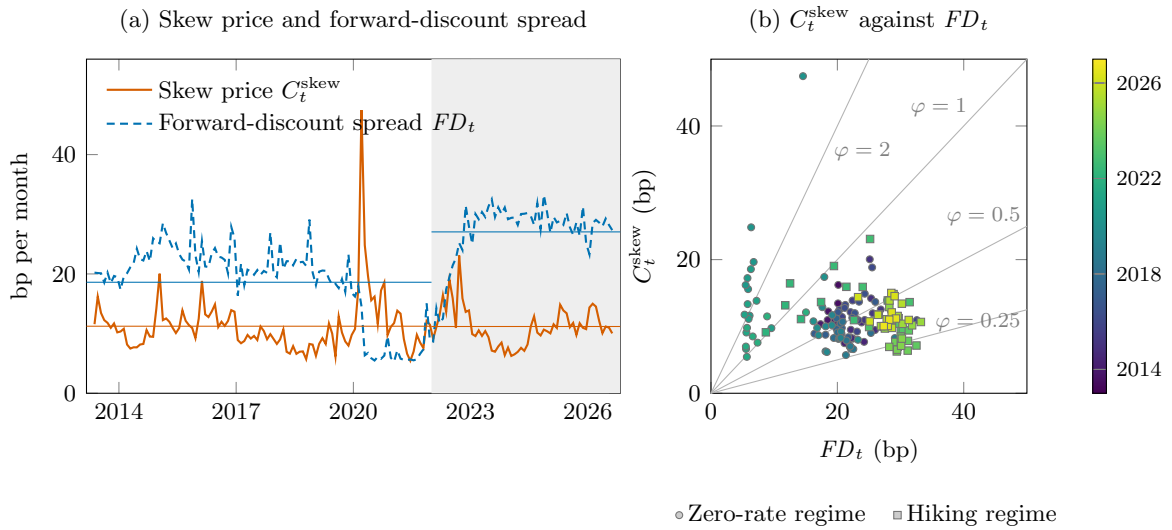


Figure 1: Skew price of protection and forward-discount spread. Panel (a) plots the skew price C_t^{skew} of one-month 10Δ protection and the forward-discount spread FD_t of the carry portfolio at each month-end of the primary sample, May 2013 to August 2026, in bp per month; thin horizontal lines mark the regime means and the hiking regime is shaded. Panel (b) plots C_t^{skew} against FD_t for the same 160 month-ends, coloured by year; circles are zero-rate and squares hiking month-ends, and the grey rays through the origin are lines of constant $\varphi_t = C_t^{\text{skew}}/FD_t$. Portfolio-level series as published in the repository. Source: LSEG Workspace; my calculations.

the grouping variable. Its 95% test-inversion set, constructed as for θ_{UB} and, with one binary instrument, an Anderson and Rubin (1949) set with a Newey–West variance, is $[-0.26, 0.49]$, and $[-0.23, 0.48]$ with a stationary bootstrap that resamples the pair jointly; it is bounded because ΔFD is significant. At most about half of the rise in the spread was matched by a rise in the skew price, and a one-for-one rise is rejected ($t = -2.66$, bootstrap $p = 0.009$).

In the rare-disaster model of Farhi and Gabaix (2016), a pair's risk reversal is proportional to the difference in the two countries' resilience (Lemma 2, eq. (34)), the disaster-related part of the interest differential and the expected carry return move with that difference (Proposition 3, eq. (22); Proposition 4, eqs. (24)–(25)), and inflation moves nominal rates without moving risk premia (proof of Proposition 9); that risk reversals are then unmoved as well is my inference from Lemma 2. Applied to a change between regimes, which they do not consider, their equations say that a wider spread with unchanged risk reversals and skew prices did not come from disaster exposure but from a component of the differential that carries no premium, so that the expected carry return need not have risen. This reading is mine, and it inherits the random-walk interpretation of the spread that underlies φ: carry equals the expected return only if the expected spot change is zero (Koijen, Moskowitz, Pedersen and Vrugt, 2018, eqs. (4)–(5)), which disaster models reject. The evidence extends to 2013–2026 the weak within-currency relation between option skew and interest differentials found earlier (Brunnermeier, Nagel and Pedersen, 2008, Table 4 of their working paper; Jurek, 2014, footnote 12 of the 2013 version), against the significant relation for option-implied disaster exposure in Farhi et al. (2015, Table 5, Panel A).

The extended sample gives secondary evidence only, because Fenics butterflies before 2010 are often stale. Over January 2007 to April 2013 (72 month-ends with at least four currencies, Fenics quotes, two long and two short) the mean C^{skew} is 22.3 bp and the mean φ is 0.64.

Table 3: E2: predictive regressions of HML^U_{t+1}

Panel A: in sample (159 months) Predictor	b	HAC t	Bootstrap bias	Analytic bias	Corrected b	One-sided p
φ	0.0062	5.95	0.00055	0.00049	0.0057	0.021
Oriented 10Δ RR	0.397	4.13	0.0183	0.0148	0.379	0.004

Panel B: out of sample (116 forecasts from January 2017) Predictor	OOS R^2	Clark–West t (p)	Clark–West t, Newey–West (p)
φ	3.7%	1.639 (0.051)	1.50 (0.067)
Oriented 10Δ RR	−5.5%	−1.11 (0.87)	−1.18 (0.88)

The dependent variable is the monthly return in decimal fractions; φ is dimensionless and the risk reversal is in decimal volatility units. HAC t is for the uncorrected b. Bootstrap bias from the residual bootstrap under the null (9,999 draws); analytic bias from Stambaugh (1999, eq. (18)). The corrected b subtracts the bootstrap bias; the one-sided p-value, for $b > 0$, is from the same bootstrap. Out of sample: expanding window with training from the extended sample, one-sided test against the historical mean with the least-squares standard error; Newey–West alongside.

5.2 E2: predictability of carry returns

Table 3 reports E2.[13] The two predictors are persistent (AR(1) coefficients of 0.86 for φ and 0.52 for the risk reversal), with a correlation of 0.56, and their innovations are negatively correlated with returns (−0.39 and −0.53), so the least-squares slope is biased upwards in small samples (Stambaugh, 1999). After the bias correction, a higher skew price per unit of forward discount predicts higher carry returns in the following month (slope 0.0057, one-sided $p = 0.021$), the sign a crash-compensation reading implies. The Newey–West t-statistic of 5.95 on the uncorrected slope is far stronger than the bootstrap evidence; for a persistent predictor with a heavy right tail it is unreliable, and it is not the pre-registered test.

Out of sample, the Clark–West statistic of 1.639 does not reach the one-sided 5% critical value of 1.645 ($p = 0.051$; 1.50 with the Newey–West standard error), so predictability is not established. Because training starts with the extended sample, the test splices the Fenics φ of 2007–2013, built from two long and two short legs, with the composite primary series built from three and three; training from the primary start gives 0.96 (Section 6). Figure 2 plots the cumulative difference between the squared errors of the two forecasts, the graphical diagnostic of Welch and Goyal (2008). The gain is concentrated in one episode: forecasts of returns in 2020, when forward discounts were smallest and φ largest, account for 89% of the cumulative difference in squared errors and for 79% of the numerator of the Clark–West statistic. Without them the statistic is 1.24 (104 forecasts), and from January 2022 the regression has forecast slightly worse than the historical mean (post hoc). The risk reversal predicts in sample (corrected slope 0.379, $p = 0.004$) but not out of sample.

The design also names option-implied variance and skewness as secondary predictors. Their moments are those of $\log(X_T/F_X)$ under the USD one-month forward measure, with dollar-base pairs converted by R1, from contracts spanned by out-of-the-money options (R3). Between the 10Δ strikes the contracts come from the calibrated smile by composite Simpson's rule with the forward as a node, converged to 10^{-10} relative in all 1,440 currency-months; beyond them each contract is bounded by R4, and the variance and skewness intervals are the exact ranges over the box of contract intervals. Two tail settings were fixed before estimation: (i) the local elasticities of the distribution function implied by the smile, Kg/G in the lower tail and $Kg/\bar{G}$ in the upper, at the 10Δ strikes (median $\gamma = 61$, $\eta = 67$), which assumes the tails beyond the quotes are no heavier than at the quotes; and (ii) $\gamma = \eta = 2$. The predictors are variance and skewness oriented to each leg (sign reversed for short legs), averaged over the six E1 legs.

[13] Bakshi and Panayotov (2013) find in-sample and out-of-sample predictability of carry payoffs from changes in commodity prices and currency volatility, and Cenedese, Sarno and Tsiakas (2014) find that high foreign-exchange market variance precedes large carry losses; the predictor here is the skew price of protection.

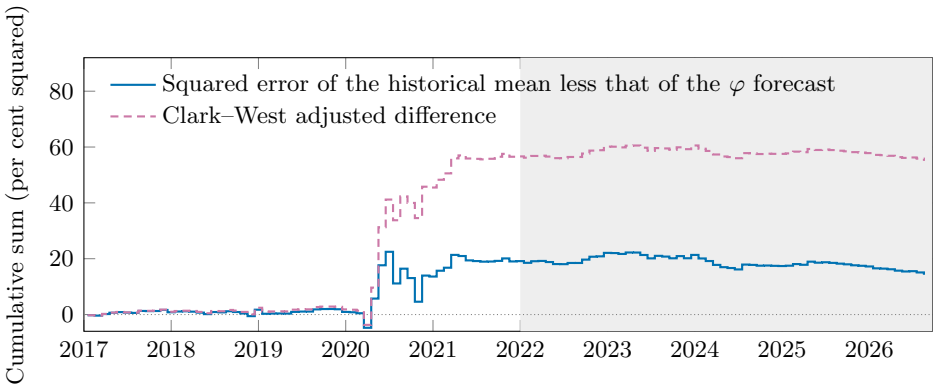


Figure 2: Out-of-sample performance of φ (post hoc). Cumulative sums over the 116 one-month forecasts from January 2017 of the squared error of the historical mean less that of the expanding-window regression on φ (solid) and of the Clark–West adjusted difference (dashed), whose mean is the numerator of the statistic of 1.639, as in the graphical diagnostic of Welch and Goyal (2008). Returns are in per cent and are plotted at the month of the return; a rising line means that the regression forecasts better; the hiking regime is shaded. Source: LSEG Workspace; my calculations.

Table 4: E2, secondary predictors: option-implied moments under tail setting (i)

Predictor	b (lower, mid, upper)	Newey–West t	One-sided bootstrap p
Variance	4.98, 4.81, 4.64	0.87, 0.90, 0.93	0.186, 0.175, 0.167 ($b > 0$)
Oriented skewness	−0.024, −0.028, −0.028	−2.74, −2.59, −1.87	0.013, 0.014, 0.037 ($b < 0$)

HML^{U}_{t+1}, in decimal fractions, on the predictor evaluated at the lower endpoint, midpoint and upper endpoint of its R4 interval, 159 months. Variance is that of $\log(X_T/F_X)$; skewness is dimensionless. A conclusion is reported only if it holds at all three.

Identification is weak. Under setting (ii) no skewness interval excludes zero, and the variance interval is about fifty times wider than under (i), so truncated quotes do not identify skewness without a tail assumption of the strength of (i). Under setting (i) the median currency-month skewness interval is $[-0.70, 0.14]$, and 32% of intervals exclude zero; but the calibrated smile's own extrapolation contradicts setting (i) in about three quarters of currency-months in each tail (75.8% in the lower and 74.1% in the upper).

Table 4 reports the regressions under setting (i). Variance does not predict carry returns at any endpoint. More negative oriented skewness, meaning more left-tail risk in the carry position, predicts higher returns, the same economic direction as the test on φ. The one-sided bootstrap test rejects at 5% at all three endpoints, but the Newey–West t does not reach the two-sided 5% critical value at the upper endpoint (−1.87). The result holds across the interval only under the bootstrap and only under a tail setting that the smile's own wings contradict, and that the boundary prices themselves exclude in three quarters of currency-months (below), so it is conditional supporting evidence, not a finding.

A post hoc analysis with R10 makes the identification statement precise (Table 5; Figure 3 illustrates the breakdown point in a Heston model and plots its distribution across currency-months). Across the 1,440 currency-months the largest exponents compatible with the smile's boundary price and slope have medians $\bar{\gamma} = 59.7$ and $\bar{\eta} = 67.8$. Setting (i) exceeds them in the lower tail in 54.4% of currency-months and in one or both tails in 75.3%; its ratio $\gamma_i/\bar{\gamma}$ has median 1.01, so it sits at the strongest assumption the boundary admits. Every currency-month in which this check refutes setting (i) also fails the smile-extrapolation diagnostic above, which fails in some further months, as a necessary condition should allow. With the sharp identified set, the sign of the skewness is identified in 93.3% of the currency-months where setting (i) is

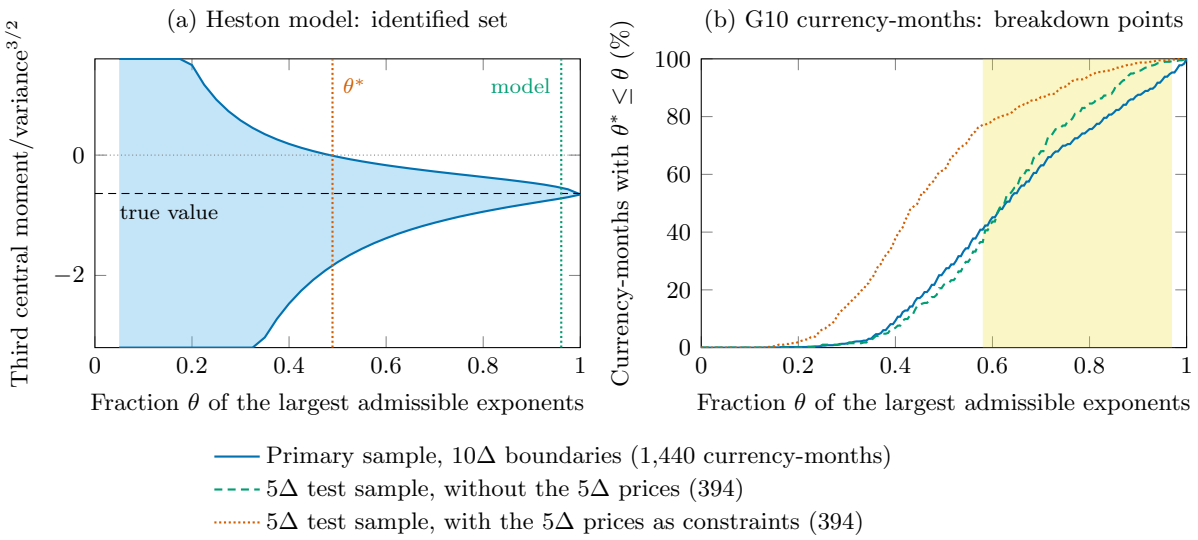


Figure 3: Identification of the sign of one-month risk-neutral skewness (post hoc). Panel (a) is synthetic: for the Heston model with $\rho = -0.3$ of Table 1 (EURUSD-type market), it plots the R10 identified set of the third central moment of $\ln(X_T/F)$, divided by the model's variance to the power 3/2, when the tail exponents beyond the 10Δ strikes are a fraction θ of the largest the boundary prices admit; the dashed line is the model's skewness, θ^* the breakdown point and 'model' the fraction implied by the model's own tails. Panel (b) plots the share of currency-months whose breakdown point θ^* is at most θ, for the primary sample and, on the 394 currency-months of the 5Δ test where θ^* is defined with them, without and with the 5Δ prices as constraints; the shaded band is the range of fractions implied by the models of Table 1. Source for panel (b): LSEG Workspace; my calculations.

admissible, but in none under setting (ii), in 26.7% at half the maximal exponents and in 70.3% at three quarters of them. The breakdown point θ^* has quartiles 0.49, 0.63 and 0.80, so in the median currency-month the sign survives only if the tail exponents beyond the 10Δ strikes are at least 63% of the largest ones the boundary admits. In the Merton and Heston models of Table 1 the models' own exponents are between 0.58 and 0.97 of the maximal ones. Truncated one-month quotes therefore identify the sign of risk-neutral skewness only under tail assumptions close to the strongest the boundary prices admit, which standard models sometimes meet and sometimes do not. At the maximal exponents, where the moments are point-identified, the skewness has median −0.34 and is negative in 78.5% of currency-months.

These hypotheses concern the tails beyond the 10Δ strikes, which the quotes used so far do not reach. The one-month 5Δ quotes of the Fenics contributor, which entered no calibration, test them out of sample. I recorded the test in the research log before retrieving their history. For each of the 450 currency-months from July 2022 to August 2026 the boundary is the Fenics smile at its 10Δ strikes, the 5Δ volatilities are $\sigma_{ATM} + BF_5 \mp RR_5/2$ (the simplified formula of Reiswich and Wystup, 2012, which reads the butterfly as a smile strangle), and the test uses a consequence of R10: under $H_L(\gamma)$ the put price at $K = kK_{\min}$ lies in $[\max(0, P_0 - G_0K_{\min}(1 - k^{\gamma+1})/(\gamma+1)), P_0k^{\gamma+1}]$, whose upper end is R4's bound and whose ends move monotonically in γ, and likewise for calls (Table 6); at the weakest hypothesis, the limit $\gamma \to 0$ (or $\eta \to 1$), the upper end is approached but not attained. The 5Δ risk reversal has the sign of the 10Δ one in 99.8% of currency-months. The class of hypotheses is consistent with the 5Δ prices in both tails in 99.1% of currency-months, all four exceptions being USDJPY months, and the largest exponents the 5Δ prices allow are high, with median fractions of the maximum of 0.88 in the lower and 0.86 in the upper tail. With the 5Δ prices as constraints and the exponents at these values, the sign of the skewness is identified in 90.6% of the 446 currency-months consistent with

Table 5: Identification of the sign of one-month risk-neutral skewness (post hoc)

Tail exponents	Admissible	Sign identified (sharp set)	Sign identified (R4 box)
Setting (ii), $\gamma = \eta = 2$	100%	0.0%	0.0%
Setting (i), boundary elasticities	24.7%	93.3%	32.1%
$\theta = 0.25$	100%	0.7%	
$\theta = 0.5$	100%	26.7%	
$\theta = 0.75$	100%	70.3%	
$\theta = 0.9$	100%	87.6%	

1,440 currency-months of the primary sample, one-month smiles, 10Δ boundaries. θ: fraction of the largest compatible exponents, $\gamma = \theta\bar{\gamma}$ and $\eta = 1 + \theta(\bar{\eta} - 1)$ (R10). Admissible: share of currency-months in which the exponents do not exceed the largest compatible ones. Sharp set: share of admissible currency-months in which the R10 identified set of the third central moment excludes zero. R4 box: share of all currency-months in which the skewness range over the box of R4 contract intervals excludes zero (Table 4). Identified sets computed on a grid of 1,000 step points per tail; a fourfold refinement in 40 currency-months moved no bound by more than 1.2×10^{-3} of its interval width. The optimisation over the mean uses a grid of means refined near the best, so the sets are inner approximations; in the same 40 currency-months, eight times as many means widened no range by more than 2.8×10^{-3} of its width and changed no identification of the sign and no θ^*.

Table 6: Out-of-sample test of the tail hypotheses with one-month 5Δ quotes (post hoc)

	Lower tail	Upper tail
5Δ price consistent with the weakest hypothesis	99.1%	99.6%
Largest consistent fraction θ_5 of the maximal exponent (quartiles)	0.82, 0.88, 0.95	0.81, 0.86, 0.93
Sign of skewness identified at the largest exponents allowed, with the 5Δ constraints	90.6% of 446	
Breakdown point with the 5Δ constraints (quartiles, where identified)	0.36, 0.45, 0.57	
Breakdown point without them, same currency-months (quartiles; supplementary)	0.53, 0.62, 0.73	

Fenics one-month quotes, July 2022 to August 2026, nine currencies, 450 currency-months. Boundary: the Fenics market-reading SABR smile at its 10Δ strikes. 5Δ volatilities from the smile-strangle reading of the quotes. The plan was recorded before the 5Δ history was retrieved; the comparison without the 5Δ prices was not in the plan.

the hypotheses. By the rule fixed in the plan, the 5Δ quotes therefore do not rule identification out in nine currency-months out of ten, and in the remaining tenth the sign is identified under no hypothesis of the class consistent with them. Consistency with one further price is necessary for a hypothesis, not sufficient, so this is the favourable case. The 5Δ prices also lower the breakdown point, from a median of 0.62 without them to 0.45 with them on the 394 currency-months where the breakdown point with them is defined (a supplementary comparison), and they lie about 4% below the SABR smile's extrapolation. The sample is short, lies in the hiking regime and consists of indications from one contributor.

5.3 E3: decomposition of the realised hedge cost

Table 7 reports E3. Unhedged carry earns 17.7 bp per month with a standard error of 12.1, so the carry premium itself is not significant in this sample ($t = 1.47$); the 10Δ hedge lowers the mean to 8.3 bp. Term (iii), −11.2 bp per month, is the skew premium paid ex ante and carried to delivery, equal to the skew price of E1 by construction; its standard error of 0.65 measures the time variation of a premium known at the start of each month, not uncertainty about realised compensation, which is all in term (i). The point estimate of the realised hedge cost is close to the skew premium because the payoff term (0.5 bp, s.e. 3.8) and the volatility-level term (1.4 bp, s.e. 1.4) are indistinguishable from zero. The total hedge cost, $HML^H - HML^U$,

Table 7: E3: realised returns and the decomposition of the hedge cost

Quantity	Mean	s.e.
HML^U (unhedged)	17.7	12.1
HML^H, 10Δ	8.3	11.9
HML^H, 25Δ	9.1	11.3
HML^H, at-the-money	9.7	8.9
(i) payoff less $\hat{\sigma}^P$ value	0.5	3.8
(ii) volatility level	1.4	1.4
(iii) skew (ex-ante premium)	-11.2	0.65

Primary sample, 159 months, bp per month, Newey–West standard errors. Terms (i)–(iii) are for the 10Δ hedge.

is -9.3 bp per month with a Newey–West standard error of 4.0 ($t = -2.35$; post hoc), but the normal reference is unreliable for it. $HML^H - HML^U$ is the net profit of the options, which is strongly right-skewed (skewness 3.6), and for the mean of right-skewed data the t-statistic has a heavier left tail than the normal (Hall, 1992). A studentised version of the stationary bootstrap, the bootstrap-t of Hall (1992) applied to dependent data with the Newey–West standard error recomputed in each of 9,999 draws, puts the 2.5% quantile of the t-statistic at -2.78 and gives a 95% interval of $[-16.4, 1.7]$ bp, which contains zero, so a two-sided test at 5% does not reject (the one-sided bootstrap p-value is 0.04). For block resampling, Götze and Künsch (1996) show that the accuracy of a studentised bootstrap depends on the variance estimator, so I claim no higher-order accuracy for this one. I therefore do not claim that the realised hedge cost differs from zero. How many large payoffs the sample contains decides the result: without the month in which the options' net profit was largest the t-statistic is -3.55, and over the extended sample, which contains 2008 but holds two legs per side priced from Fenics smiles, it is -0.25. Panel (b) of Figure 4 shows why: the realised cost follows the premium paid, apart from a few months in which the options paid. By regime (supplementary, post hoc), unhedged carry averages 7.4 bp (s.e. 12.4) before 2022 and 37.0 bp (17.8) after, and hedged carry -2.7 bp (15.7) and 29.3 bp (16.0); neither difference is significant, with bootstrap 95% intervals of $[-19.5, 80.0]$ and $[-16.8, 80.3]$ bp. The design states the decomposition for both regimes. Before 2022 and from 2022, the payoff term averages -0.4 bp (s.e. 5.3) and 2.2 bp (4.9), the volatility-level term 1.5 bp (1.9) and 1.3 bp (1.5), and the skew term -11.24 bp (0.82) and -11.21 bp (0.84). No term differs between the regimes: the bootstrap 95% intervals of the differences, with 9,999 draws resampling each regime separately as for E1, are $[-13.2, 18.9]$, $[-5.4, 5.1]$ and $[-2.2, 2.4]$ bp. The cost of 10Δ protection is therefore close to its ex-ante skew premium in both regimes, and neither its realised nor its ex-ante part shows a change between them.

θ_{UB} is 0.53 at 10Δ, 0.49 at 25Δ and 0.45 at the money, against a diffusive null of $\theta_0 \approx 0.10$ at 10Δ (the average forward delta of the hedges) and about 0.25 for a 25Δ put by R6. Its 95% test-inversion set is unbounded, because the mean unhedged return is not significantly different from zero; as the design anticipated, the ratio is weakly identified, no value is claimed, and the gap between 0.53 and 0.10 is not read as a crash share. At 10Δ the set is the line less the interval from -1.23 to 0.09 (post hoc): it excludes $\vartheta = 0$ only under the normal reference questioned above, and it retains θ_0 narrowly, with $t = -1.93$ at the automatic lag of six and rejection at every fixed lag up to five. The delta-method intervals that the design names as secondary are $[-0.21, 1.27]$ at 10Δ, $[-0.30, 1.27]$ at 25Δ and $[-0.30, 1.20]$ at the money. They are bounded only because the delta method holds the standard error at its value at θ_{UB} while dividing by a mean that cannot be distinguished from zero, and procedures that always return bounded sets have zero worst-case coverage for such ratios (R9(c)). The realised terms also depend on the crashes that happened to occur in the sample, a peso-problem caveat: in the extended sample (secondary; Fenics quotes, 72 months, 2007–2013) term (i) is 16.7 bp (s.e. 10.2), reflecting option

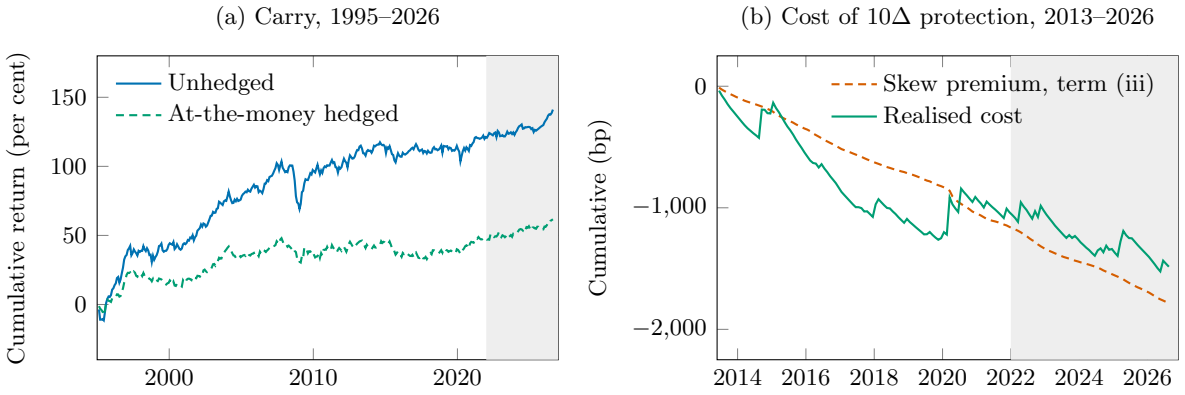


Figure 4: Carry returns and the cost of crash protection. Panel (a) plots cumulative unhedged and at-the-money-hedged carry returns of the long sample, January 1995 to July 2026, as simple sums of monthly returns in per cent. Panel (b) plots, for the primary sample, the cumulative realised cost of 10Δ protection, $HML^H - HML^U$, and the cumulative skew term (iii), the premium paid ex ante, in bp. Returns are plotted at the month of the return; the hiking regime is shaded. Portfolio-level series as published in the repository. Source: LSEG Workspace; my calculations.

payoffs in 2008, the skew term is −22.3 bp (s.e. 3.0) and the unhedged mean is 8.4 bp with a standard error of 62.7.

The long at-the-money sample (Section 3; panel (a) of Figure 4) asks what a longer sample identifies. From January 1995 to July 2026 (379 months) unhedged carry earns 37.2 bp per month (s.e. 13.2, $t = 2.83$) and at-the-money-hedged carry 16.3 bp (s.e. 7.7), so the hedge gives up 20.9 bp (s.e. 7.9). With a significant unhedged premium the test-inversion set of $\theta_{\text{UB}} = 0.56$ is bounded, unlike the primary sample's: $[0.27, 0.94]$, or $[0.33, 0.85]$ without 2008. For an at-the-money hedge the diffusive null is about one half (0.50, whether the dollar-base legs use the pips or the premium-adjusted forward delta), and the set contains it, as does every sub-period's set; the at-the-money hedge gives up about what a diffusion predicts. It has no skew term, and the set is wide, so the result is uninformative about priced crash risk rather than evidence against it. In 2008 the hedge paid: the option payoff averaged 502 bp per month against a premium of 378 bp, and hedged carry lost 99 bp per month against 222 bp unhedged, so the positive θ_{UB} of that year measures a gain, not a cost share. The sample shows what bounding the ratio at 10Δ would take: the set is bounded when the t-statistic of the unhedged mean exceeds 1.96, and at the primary sample's ratio of mean to long-run standard deviation that needs about $159 \times (1.96/1.47)^2 \approx 280$ months, some 24 years, of 10Δ-hedged returns. The one-month risk-reversal quotes that such a hedge needs begin in 2007 (Fenics) and between 2010 and 2013 (composite) in the licensed panel, against 1995 for the at-the-money volatility.

The design compares θ_{UB} only with premium shares in the literature, noting differences in hedge moneyness. Jurek (2014) compares hedged and unhedged G10 carry over 1999–2012 with out-of-the-money options and finds that crash premia account for at most one third of the excess return. Unlike θ_{UB}, his hedged trades hedge the delta of the option overlay and so earn about the whole diffusive premium; he reads his share as an upper bound because peso problems bias the unhedged mean upwards (Section 2.2 of the August 2013 version). On his decomposition the comparable quantity here is $(\theta_{\text{UB}} - \theta_0)/(1 - \theta_0)$, about 0.48 at 10Δ; its set is unbounded, like that of θ_{UB}, and contains one third, so the two do not conflict. Daniel, Hodrick and Lu (2017) use the same unlevered construction as this paper, one option per unit of forward with the premium borrowed, with 10Δ and 25Δ options over 2000–2013 and without SEK; in their Table 11 the hedges give up about 6 to 17 per cent of the unhedged mean at 10Δ and 23 to 39 per cent at 25Δ, close to the diffusive nulls of 0.10 and 0.25. Farhi et al. (2015) estimate from a structural model, not from a hedge, that disaster risk accounts for more than a third of currency risk premia in advanced countries over 1996–2011 (NBER revision of May 2013, abstract); no

Table 8: E5: exposures to VIX roll-down and FX-volatility innovations

Portfolio	α (bp)	β, VIX roll-down	β, FX-volatility innovation	R^2
HML^U	-7.3 (-0.61)	0.049 (7.41)	-3.11 (-2.16)	0.33
HML^H, 10Δ	-10.6 (-0.79)	0.039 (4.69)	-1.76 (-1.21)	0.22

Primary sample, 159 months. Newey–West t-statistics in parentheses. Returns are in decimal fractions and α in bp per month; the VIX roll-down is a decimal return and the FX-volatility innovation is in units of mean absolute daily log change. The AR(1) coefficient of the FX-volatility level is 0.76.

hedge moneyness enters their share, and θ_{UB} is not a model share of the premium. Close to the money, the comparison is with Burnside et al. (2011): over February 1987 to April 2009 and the six currencies of their option data, their hedged carry trade earns 1.58 per cent a year against 2.96 per cent unhedged (NBER Working Paper 14054, Section 5 and Table 5), a hedge-cost ratio $1 - 1.58/2.96$ of about 0.47. That is close to the at-the-money θ_{UB} of 0.45 in the primary sample and 0.56 in the long sample, and to the at-the-money diffusive null of one half: near the money, the hedge gives up about what a diffusion predicts in their sample as in mine.[14]

5.4 E5: systemic-risk exposure

Table 8 reports E5. Unhedged carry returns load strongly on the VIX roll-down (correlation 0.55), consistent with Caballero and Doyle (2012), and negatively on FX-volatility innovations. The 10Δ hedge lowers the VIX loading by about a fifth but leaves most of it, whereas Caballero and Doyle (2012) report that option-hedged carry retains high returns that systemic risk does not account for; the hedged FX-volatility loading is not significant. After adjusting for both factors, neither portfolio earns a significant α; the exposure-adjusted ratio $1 - \alpha_H/\alpha_U$ is -0.45 and its test-inversion set is unbounded, so it is unidentified. In the extended sample (secondary, 65 months) the loadings are larger, 0.124 unhedged ($t = 5.39$) and 0.112 hedged ($t = 4.53$), and the αs are not significant.

The design also asks for a descriptive account of smiles, φ_t and returns around 5 August 2024, which I computed after the main results from daily composite quotes between 15 July and 30 August 2024, calibrating a smile for every currency on every New York business day under the month-end rules; all 315 smiles converged and passed both static-arbitrage checks. I report only averages over at least six currencies. Averaged over the nine currencies, the one-month at-the-money volatility rose from 7.98 volatility points on 31 July to 10.27 on 5 August, and the absolute 25Δ risk reversal from 0.73 to 1.15, both their highest values in the window; by 30 August they were 8.72 and 0.66. For one-month 10Δ protection on the portfolio formed on each day as at a month-end, the skew price rose from 11.2 bp on 31 July to 19.4 bp on 5 August while the forward-discount spread barely moved (29.7 and 31.0 bp), so φ_t rose from 0.38 to 0.63, its highest value in the window, against 0.25 to 0.40 in the second half of July; it was 0.46 on 30 August. Marked at each day's spot against its forward, the position opened at the 31 July month-end was worth -310 bp of notional on 5 August, its low, against $+30$ bp on 31 July, when the mark equals the forward discount it earns; its 10Δ options were still out of the money at that day's spot, so at intrinsic value they offset none of the loss. The position recovered by the end of August, and over the month to expiry it earned 25.8 bp unhedged and 42.6 bp hedged. The price of crash insurance rose sharply with volatility in the episode and fell most of the way back within the month, while the forward-discount spread hardly moved.

[14] Lempérière, Deremble, Nguyen, Seager, Potters and Bouchaud (2017) report, for a pool of 20 developed-country currencies, that they could not confirm Jurek's result and that hedging with at-the-money options removes the carry profit. In my long sample the at-the-money hedge gives up about half the premium, and the set $[0.27, 0.94]$ excludes one, the value at which the hedged premium vanishes.

Table 9: Robustness grid (one-month tenor)

	E1 and E2				Returns and E3 (bp per month)			
Variant	$\Delta\varphi$ (s.e.)	Bootstrap 95%	b (p)	CW	HML^U	HML^H	Skew	θ_{UB}
Base	−0.355 (0.204)	[−0.81, 0.01]	0.0057 (0.021)	0.96	17.7 (1.47)	8.3 (0.70)	−11.2	0.53
25Δ hedge	−0.272 (0.166)	[−0.64, 0.03]	0.0073 (0.015)	0.75	17.7 (1.47)	9.1 (0.81)	−8.4	0.49
ATM hedge	n.d.				17.7 (1.47)	9.7 (1.09)	0	0.45
Smile strangle	−0.345 (0.196)	[−0.79, 0.01]	0.0058 (0.022)	0.99	17.7 (1.47)	8.7 (0.73)	−10.8	0.51
Fenics quotes	−0.347 (0.203)	[−0.80, 0.02]	0.0057 (0.023)	0.98	17.7 (1.47)	8.5 (0.71)	−11.2	0.52
Stale BF missing	−0.440 (0.265)	[−1.06, 0.04]	0.0037 (0.025)	0.19	17.7 (1.53)	8.8 (0.76)	−11.2	0.50
Excl. March 2020	−0.329 (0.193)	[−0.76, 0.02]	0.0053 (0.036)	1.31	19.9 (1.73)	8.8 (0.77)	−11.0	0.56
Excl. CHF 2014–15	−0.353 (0.204)	[−0.81, 0.02]	0.0055 (0.019)	0.84	20.4 (1.86)	10.4 (0.89)	−11.2	0.49
Implem., $k = 1$	base	base	0.0058 (0.020)	0.92	5.4 (0.45)	−6.7 (−0.55)	−11.2	n.m.
Implem., $k = 1.5$	base	base	0.0058 (0.020)	0.92	5.4 (0.45)	−8.0 (−0.66)	−11.2	n.m.
Implem., $k = 2$	base	base	0.0058 (0.020)	0.92	5.4 (0.45)	−9.4 (−0.78)	−11.2	n.m.
Dollar carry	−1.060 (0.559)	[−2.19, −0.37]	0.0001 (0.397)	−1.51	0.3 (0.02)	−2.6 (−0.17)	−2.8	n.m.
Ten currencies	−0.300 (0.149)	[−0.64, −0.02]	0.0051 (0.058)	1.17	15.3 (1.37)	8.2 (0.74)	−9.7	0.47
Log returns	base	base	0.0055 (0.023)	0.86	16.7 (1.38)	7.4 (0.62)	−11.2	0.56
Previous-day spot	base	base	base	base	17.7 (1.47)	5.2 (0.43)	−11.2	0.70
Vanna–volga	−0.337 (0.190)	[−0.77, 0.01]	0.0059 (0.024)	1.08	17.7 (1.47)	8.7 (0.73)	−10.8	0.51

Primary sample; 160 month-ends for φ and 159 return months, except the at-the-money (ATM) hedge (159), stale butterflies (BF) missing (156, 155) and excluding March 2020 (the February and March 2020 month-ends; 158, 157); CHF is excluded at the November 2014 to March 2015 month-ends. $\Delta\varphi$: hiking minus zero-rate, Newey–West s.e. in parentheses, stationary-bootstrap 95% interval. b: bias-corrected E2 slope on φ, one-sided bootstrap p. CW: Clark–West statistic with training from the primary start (116 forecasts; 112 with stale butterflies missing and 114 excluding March 2020), not comparable with 1.639 in Table 3 (training from 2007). θ_{UB}: point estimate; its test-inversion set is unbounded in every row. Returns: Newey–West t in parentheses. Implem.: implementable, option half-spread k times the ATM half-spread. base: as in the base row; n.d.: not defined; n.m.: not meaningful (mean unhedged return within half a standard error of zero). Bootstraps use 9,999 draws.

5.5 The regime-shift question under the pre-registered rules

The design requires three conditions: an E1 difference whose interval excludes zero (not met: −0.355, bootstrap interval $[-0.814, 0.014]$), a bias-corrected $b > 0$ in E2 (met: 0.0057, one-sided $p = 0.021$) and an out-of-sample Clark–West rejection at 5% (not met: 1.639 against 1.645). Only the second holds, so the claim that the ex-ante price of crash insurance explains the change in the carry premium between the regimes has partial support only, the pre-registered label for this outcome: the condition met concerns in-sample predictability, and no estimate indicates that the price of crash insurance changed between the regimes. The skew-term rule of Section 4 is uninformative here: term (iii) is an ex-ante premium, whose interval excludes zero whenever smiles are skewed, so its precision is mechanical and is not by itself evidence of priced crash risk.

6 Robustness

6.1 The one-month grid

Table 9 reports the robustness grid. The rules the design leaves open were fixed in the estimation script after the primary results were recorded but before any variant was estimated. A currency is dropped at a month-end if its selected 25Δ butterfly lies in a run of five or more identical two-sided business-day quotes (109 currency-months). A month-end is removed if its quote date or return window falls in an excluded period. Dollar carry follows Lustig, Roussanov and Verdelhan (2014, Section 3.2), long or short all available currencies with equal weights by the sign of their average forward discount (here the nine-currency average). In the ten-currency ranking the dollar enters with $fd = 0$ and a dollar leg has zero return and no option; the previous-day variant changes option payoffs only.

φ is lower in the hiking regime in every variant. The difference is significant at 5% under both inferences only in the ten-currency ranking ($t = -2.01$, interval $[-0.64, -0.02]$). For dollar carry only the bootstrap interval excludes zero, and its φ divides by the absolute average forward discount, below 5 bp in 40 zero-rate months, so the difference comes mainly from the denominator. In the base and ten-currency portfolios the skew price per month is nearly the same in both regimes (11.2 and 11.2 bp; 10.0 and 9.3 bp), and the fall in φ comes from the forward-discount spread. Under my reading of the rule the E1 conclusion therefore depends on the portfolio definition, although the ten-currency ranking is one of many correlated variants, with no multiplicity adjustment. The smile-strangle reading and Fenics quotes move φ by less than 4 per cent.

In E2, the bias-corrected slope is positive with one-sided $p \leq 0.036$ in every HML variant, $p = 0.058$ for ten currencies, and for dollar carry it is essentially zero (0.0001, $p = 0.40$). No variant rejects out of sample at 5% when training starts in 2013. The design fixes initial training on the extended sample for E2; for the variants I fixed the primary start before estimating any of them, because several variants cannot be built on the extended sample, and this departure was not recorded as such until 3 October (Section 4.7).

The skew term of E3 is -8.4 to -11.2 bp per month across the HML variants, and its small standard error reflects the stability of an ex-ante premium; θ_{UB} lies between 0.45 and 0.70 at mid prices, its test-inversion set is unbounded in every variant, because no unhedged mean is significant, and the hedged mean is insignificant everywhere. Quoted spreads on forwards and spot reduce the unhedged mean from 17.7 to 5.4 bp, and option spreads make the hedged mean negative for every k. With the previous day's spot the hedged mean falls by 3.1 bp, from 43 months with an option in the money on one of the two days, no single month contributing more than 0.5 bp; neither close is the 10:00 New York cut, which lies between them, so both are reported.

The research log of 23 September applies the stale-butterfly rule to the extended sample as well, which the grid does not cover. Under the audit rule the Fenics 25Δ butterflies are stale in 311 currency-months, in 67 of the 76 extended month-ends (63 of the 72 used), so treating them as missing leaves 28 month-ends with at least four currencies. Over those, the mean skew price is 16.2 bp and the mean φ 0.55, against 22.3 bp and 0.64 over all 72, and the Clark–West statistic of E2, whose training starts in the extended sample, is 1.35 instead of 1.639; out-of-sample predictability is not established either way.

6.2 Vanna–volga smiles

The alternative smile is the implied volatility of the vanna–volga price of Castagna and Mercurio (2007, eqs. (6)–(7)), with Greeks at the at-the-money volatility; pillars at the delta-neutral-straddle strike and at the 25Δ strikes at the pillar volatilities (Castagna and Mercurio, 2006, Section 2; Bossens, Rayée, Skantzos and Deelstra, 2010, Sections 3.2 and 3.3); and, under the market reading, a smile strangle solved so that the smile prices the market strangle (Reiswich, 2010, Section 3.3.3; Bossens et al., 2010, Section 3.3). I chose the price smile before estimation, because the properties Castagna and Mercurio prove belong to the price; the first-order approximation overvalues the wings (Castagna and Mercurio, 2007, p. 10) and the second-order one can be undefined (Reiswich, 2010, Section 3.3.2). An implementation rule not from a source (a 10Δ strike must connect to the at-the-money pillar through strikes where the smile is defined) does not bind. The smile passes the static-arbitrage check over ± 4 at-the-money standard deviations in 99.7% of the 1,440 currency-months (market reading), and no leg uses one that fails. It gives φ within 5 per cent of SABR in each regime, the same conclusions, and a better fit to the held-out 10Δ risk reversal and butterfly (mean absolute errors of 0.10 and 0.13 volatility points, against 0.13 and 0.16 for SABR).

Table 10: Three-month tenor, quarterly non-overlapping positions

Phase	Quarters	$\Delta\varphi$ (s.e.)	Bootstrap 95%	b (p)	HML^U	HML^H	Skew	θ_{UB}
Mar (primary)	53; 35, 18	−0.301 (0.166)	[−0.63, −0.05]	0.0222 (0.045)	47.5 (1.29)	36.6 (1.12)	−24.9	0.23
Jan	53; 34, 19	−0.275 (0.167)	[−0.62, −0.02]	0.0139 (0.127)	57.8 (1.64)	42.8 (1.25)	−23.6	0.26
Feb	54; 35, 19	−0.246 (0.130)	[−0.51, −0.04]	0.0119 (0.174)	42.9 (1.18)	9.7 (0.26)	−22.7	0.77

Phases by first quarter-end month: March, June, September, December (primary, from June 2013); January, April, July, October; February, May, August, November (both supplementary). Quarters of φ: total; zero-rate, hiking. $\Delta\varphi$ and b as in Table 9, with the residual bootstrap under the null for p. Returns and skew term in bp per quarter, Newey–West t in parentheses. Bootstraps use 9,999 draws.

6.3 Three-month tenor

Positions are quarterly and non-overlapping, three long and three short on the three-month forward discount, protected at the 10Δ strike of the three-month smile. The primary rebalancing dates are the calendar-quarter month-ends from June 2013; the other two phases are supplementary. The quotes are complete for all nine currencies from May 2013 (11 of 11,520 fields substituted, 2 missing), and 1,436 of 1,440 smiles converge under each reading, all passing both arbitrage checks. In Table 10, φ falls from about 0.6 to 0.3 in every phase; the bootstrap interval excludes zero in all three, but the Newey–West t-statistic lies between −1.65 and −1.89, so under my reading the difference is not significant. The skew price per quarter changes little (25.4 to 23.6 bp in the primary phase) while the forward-discount spread rises from 55 to 82 bp. The bias-corrected E2 slope is positive in every phase, significant at 5% only in the primary one ($p = 0.045$); the skew term (−23 to −25 bp per quarter) is stable over time (t between −14 and −24), which reflects the stability of an ex-ante premium rather than precision about realised compensation, and θ_{UB} (0.23 to 0.77) has an unbounded set in each phase. As at one month, the payoff and volatility-level terms are indistinguishable from zero: 10.0 bp (s.e. 16.1) and 4.1 bp (4.3) per quarter in the primary phase, against a skew term of −24.9 bp, with θ_0 of 0.10.

7 Conclusion

Under the pre-registered rules the answer is partial support only. The point estimate of φ falls from 0.809 to 0.454, but the interval for the difference includes zero, and a supplementary (post hoc) split shows no evidence that the skew price per month changed (11.24 and 11.19 bp). φ predicts carry returns in sample ($p = 0.021$) but not out of sample (1.639 against 1.645, with about four fifths of the statistic's numerator from forecasts of 2020): one condition of three, and it concerns predictability, not the change between regimes. Under my reading of the E1 rule the conclusion depends on the portfolio definition, but in the ten-currency ranking, the only variant where both intervals exclude zero, the fall also comes from the forward-discount spread. The evidence points to a rise in the forward-discount spread rather than a change in the price of crash insurance.

The mean realised cost of 10Δ protection is close to the skew premium paid ex ante, in both regimes, and unhedged carry is strongly exposed to the VIX roll-down, which the hedge lowers by about a fifth. The evidence cannot say what share of the carry premium compensates for crashes, or how the entropy difference behind it divides across cumulant orders (R7): the confidence sets of θ_{UB} (one and three months) and of the exposure-adjusted ratio are unbounded, because the unhedged carry premium is insignificant ($t = 1.47$ at one month, 1.29 in the primary three-month phase), and quoted spreads reduce it from 17.7 to 5.4 bp. Over 1995–2026 the premium is significant and the at-the-money set is bounded, but it contains the at-the-money diffusive null, which is also close to the ratio implied by the near-the-money hedge of Burnside et al. (2011), and an at-the-money hedge has no skew term.

Two post hoc analyses sharpen these conclusions. The skew price did not rise with the

forward-discount spread: the ratio of the two regime changes is -0.006 with a 95% set of $[-0.26, 0.49]$, and a one-for-one rise is rejected. In the model of Farhi and Gabaix (2016) this places the rise in the spread outside disaster exposure, in a component that carries no premium; that reading is mine and rests on the random-walk interpretation of the spread. The second concerns what truncated quotes can say about crash risk at all. Under the tail hypotheses of R4, the quotes admit exponents only up to bounds set by the strike elasticities of the boundary prices, the boundary-elasticity setting exceeds them in three quarters of currency-months, and the sign of one-month risk-neutral skewness is identified only if the tail exponents reach at least 63% of the admissible maximum in the median currency-month, along a common path for both tails. Standard jump-diffusion and stochastic-volatility models sometimes meet that requirement and sometimes do not, so option-implied skewness from quotes that stop at 10Δ is identified only under an explicit and strong tail assumption. Held-out 5Δ quotes, available from 2022, do not contradict that assumption: they are consistent with the tail hypotheses in 99% of currency-months, allow exponents of about 0.87 of the maximum, above the breakdown point, and do not rule identification out: under the strongest hypothesis they allow, the sign is identified in 91% of the consistent currency-months. That is the favourable case, since one further price cannot confirm a hypothesis about a whole tail.

Several limitations qualify these conclusions. The hiking regime is short (56 month-ends), and a supplementary (post hoc) analysis finds roughly 8 effective observations per regime, so the pre-registered intervals are likely too narrow. The persistence and heavy right tail of φ make the Newey–West inference in E2 unreliable, the out-of-sample result is sensitive to the training start, and the cross-section is small (nine currencies, three per side). θ_{UB} is weakly identified, and the realised E3 terms depend on the crashes that happened to occur. The 10Δ premium is extrapolated from 25Δ quotes, and SABR fits the held-out 10Δ quotes less well than vanna–volga. The sign of option-implied skewness is identified only under strong tail assumptions, and the pre-registered setting (i) is contradicted by the smile's own wings in about three quarters of currency-months. The design does not test the accounts of the instability of carry that do not rest on crash risk (Engel et al., 2022; Andrews et al., 2024). The quotes are dealer-contributed composites, not transaction prices, and implementable returns rest on an assumed spread rule. Payoffs use end-of-day spot, not the 10:00 New York cut; the hedged mean moves by 3.1 bp between the closes that bracket it. Dates use New York holidays only, and the revision check covers only the three days between my two retrievals.

Further data would narrow the remaining uncertainty. A longer hiking regime would add effective observations; executable single-option quotes would replace the spread rule; fixings at the 10:00 New York cut would settle payoff timing; quotes beyond 10Δ would narrow the R4 intervals. Above all, a bounded set for θ_{UB} at 10Δ needs a significant unhedged carry premium, about 24 years of hedged returns at the primary sample's ratio of mean to standard deviation; the long at-the-money sample shows that three decades of carry deliver one, but the risk-reversal quotes that out-of-the-money protection needs begin in 2007 in the licensed panel.

A Theory results

Table 11 lists the results used in the paper. Proofs are in `theory/notes.tex`, with the hypotheses stated in full. Status: P, proved in the notes, with the part I claim as my own named in the last column; P*, a known result whose published or standard proof the notes write out; C, cited with its hypotheses. Where a part is claimed as mine, I have not found it in this form in the sources I consulted.

Table 11: Theory results R1 to R10

	Statement	Status	Source
R1	If $Z_t = S_t B_t^f/(S_0 B_t^d)$ is a true martingale, $d\mathbb{Q}^f/d\mathbb{Q}^d = Z_T$ defines an equivalent foreign measure; Bayes formula; foreign and domestic martingale prices correspond; under Garman–Kohlhagen $\mathbb{E}^{\mathbb{Q}^d} S_T = F$ and $\mathbb{E}^{\mathbb{Q}^f}(1/S_T) = 1/F$	P*	Geman, El Karoui and Rochet (1995); Shreve (2004); strict local martingales: Carr, Fisher and Ruf (2014)
R2	A function is the call-price curve of some $X \geq 0$ with mean F if and only if it is convex, equals F at zero, has right slope at least -1 at zero and vanishes at infinity; the law is then unique	P*	Föllmer and Schied (2004, Lemma 7.23); context: Breeden and Litzenberger (1978); density test: Gatheral and Jacquier (2014); finitely many strikes: Davis and Hobson (2007)
R3	Spanning of $\mathbb{E}f(X)$ by the forward and out-of-the-money puts and calls, under an explicit integrability condition	P*	Carr and Madan (2001), standard proof; contracts of Bakshi, Kapadia and Madan (2003), centred here at the forward
R4	Bounds on the unobserved tails of put and call prices from observed wing prices and tail exponents; variance and skewness as intervals	P	Mine: the inequality under the monotone-ratio hypotheses (by the monotone form of l'Hôpital's rule) and its use for the moment contracts. Equality case (exact power-law tails): Taleb et al. (2023) in the upper tail, Benaim, Dodgson and Kainth (2008) in the lower; tail condition as a star order: Lando, Arab and Oliveira (2023); truncation bounds for model-free variance: Jiang and Tian (2005, Proposition 2); wings and finite moments: Lee (2004a); contracts adapted from Bakshi, Kapadia and Madan (2003)
R5	(a) Closed-form strike for pips spot delta; (b) the premium-adjusted call delta rises to a unique maximum then falls, the convention taking the right root; (c) a slope condition makes the pips (spot or forward) smile delta strictly monotone, so inversion is unique	P*	Reiswich and Wystup (2012) for definitions, the closed form, the convention, the non-monotonicity in (b) and the first-order condition; (b): Jäckel (2020), Mills-ratio monotonicity: Bagnoli and Bergstrom (2005), Sampford (1953); (c): the slope inequality is Fukasawa (2012, Lemma 2.1), used here as a hypothesis in two-sided form; unique strike: Mingone (2023); conditions imposed on smiles not assumed free of arbitrage: Martini and Mingone (2022); non-monotone smile deltas: Reiswich (2010), Healy (2025)

Table 11: Theory results R1 to R10 (continued)

	Statement	Status	Source
R6	Diffusive null of the hedge cost: $\theta_0 = \Phi(-d_+(\xi))$ exactly for a forward ξ between F and $Fe^{\lambda\tau}$, with $d_+(G) = [\ln(G/K) + s^2/2]/s$ for the put's strike K; within $(2\pi)^{-1/2}\lvert\lambda\rvert\sqrt{\tau}/\sigma$ of $\Phi(-d_1)$, $d_1 = d_+(F)$	P	Mine: the mean-value form and the error bound. Leading-order form: Jurek (2014, Section 2.2), Farhi et al. (2015, Section 5.1)
R7	Expected log currency excess return equals the entropy difference less the covered-interest-parity basis; cumulant expansion of entropy; entropy bound	P*	Backus, Foresi and Telmer (2001, Proposition 1 and eqs. (9)–(13)); wedge: Lustig and Verdelhan (2019); convenience yields: Jiang, Krishnamurthy and Lustig (2021); conditional entropy and the bound: Backus, Chernov and Zin (2014), who relate the bound to Bansal and Lehmann (1997) and Alvarez and Jermann (2005); sign against Du, Tepper and Verdelhan (2018); the basis step is elementary
R8	Hagan volatility with $\beta = 1$ is smooth; the calibration has a locally unique C^1 solution in the quotes under non-degeneracy conditions	P	Hagan et al. (2002); calibration conditions: Reiswich and Wystup (2012), Reiswich (2010, Section 3.2); implicit-function-theorem argument: Reiswich (2010, Theorems 2–3), Henrard (2011); numerical calibration under FX conventions: Clark (2011, Section 3.10); lemma: Hagan et al. (2002, eq. (2.18)), Choi and Wu (2021), Milnor (1963); informal identification with β fixed: Hagan et al. (2002, Section 3.1); ill-posedness with β free: Reiswich (2010, Section 3.2.2); uniqueness only within SABR: Healy (2025); explicit leading-order inversion: Le Floc'h and Kennedy (2014); the joint smoothness and local well-posedness statements for SABR are mine
R9	Central limit theorem for strongly mixing sequences (strict stationarity, $2+\delta$ moments, summable $\alpha(n)^{\delta/(2+\delta)}$, positive long-run variance); validity of the stationary bootstrap; confidence sets for ratios that have finite expected diameter, or are bounded with probability one, have zero worst-case coverage, hence test inversion	C	Ibragimov (1962, Theorem 1.7); Politis and Romano (1994), Theorem 2 of the 1991 report; Gonçalves and de Jong (2003); Gleser and Hwang (1987); Dufour (1997); test inversion for a ratio: Fieller (1954)

Table 11: Theory results R1 to R10 (continued)

	Statement	Status	Source
R10	The exponents compatible with the boundary price and slope are $0 < \gamma \leq K_{\min} G_0/P_0 - 1$ and $1 < \eta \leq 1 + K_{\max} \bar{G}_0/C_0$, with unique power-law tails at the maximum; the identified set of the raw moments is the observed middle part plus two compact convex tail sets whose extreme points, without further price constraints, come from mixing measures with at most two atoms; for nonnegative weights R4's upper bound is sharp and, for weights positive near the boundary, its lower bound is not; a check of setting (i); breakdown point of the sign of skewness	P	Mine: that these exponents are the largest admissible in the whole classes, with uniqueness there; the generating family with the boundary price and slope fixed, the identified set and its consequences for R4; the check of setting (i); the breakdown analysis of skewness. Matching power-law tails and their exponents: Benaim, Dodgson and Kainth (2008), Qin et al. (2026); star order: Lando, Arab and Oliveira (2023); elasticity: Veres-Ferrer and Pavía (2022); Pareto index and convexity: Taleb et al. (2023); tail decay: Bollerslev, Todorov and Xu (2015); bounds from option prices: Bertsimas and Popescu (2002); shape-constrained tails with fixed boundary values: Lam and Mottet (2017), Mottet and Lam (2017); tails attached to recovered densities: Figlewski (2010), Bollinger, Melick and Thomas (2023); moment problem: Kemperman (1968), Winkler (1988, as quoted by Pinelis, 2016), Popescu (2005); breakdown: Horowitz and Manski (1995), Stoye (2005, 2010), Masten and Poirier (2020), Kline and Santos (2013); related bounds: Zhuang (2026), Bondarenko et al. (2026)
GK	The Garman–Kohlhagen price solves its PDE and is the unique $C^{1,2}$ solution of polynomial growth; no boundary condition at $S = 0$ is needed	P*	Garman and Kohlhagen (1983); Itô and Feynman–Kac, standard results stated for example in Shreve (2004)

B Reproduction and data access

The code, under the MIT licence, is in the GitHub repository `https://github.com/Amanjeet-S/Crash-Insurance-G10-Carry-Premium`, and commands run from its root; this paper, the theory notes and the published series are under CC BY 4.0. Without an LSEG licence, the results listed in Section 3.6 can be recomputed from the published series in `data/public/fx_carry_portfolio_series` with `.venv/bin/python scripts/reproduce_from_public.py`, in the first environment created below, which compares each recomputed number with the number printed here; the E5 factor and the comparison with published portfolios use public downloads by `scripts/acquire_cboe_vx.py` and `scripts/acquire_verdelhan.py`. The rest needs the quotes: E4 and the 25Δ skew prices, every result of the oriented risk reversal, the other robustness variants, the three-month tenor, the per-currency analyses (identification, the per-currency moment intervals, the 5Δ test, the calibration diagnostics and the design checks), the extended-sample stale-butterfly check, the account of 5 August 2024, the 2008 payoff and premium and the θ_0 of the long sample's at-the-money hedge, the check of the Cboe settlements against LSEG's continuation, the splice of Fenics and composite quotes, the revision check and the audit counts. The script lists each with its reason. The theory results, the model validation and the kernel checks are synthetic and are reproduced by the tests and `scripts/validate_moments_models.py`; the figures are written by `scripts/make_figures.py`, three of them from the published series alone. The two Python environments, which need Python 3.11 or later (I used 3.13.9), are created with

```
python3 -m venv .venv && .venv/bin/pip install -r requirements.lock \
  && .venv/bin/pip install -e . --no-deps
.venv/bin/python -m pytest -q
```

```
python3 -m venv .venv-lseg && .venv-lseg/bin/pip install -r requirements-lseg.lock
```

The LSEG acquisition reads the App Key of whoever runs it from `~/.lseg/app_key`. The steps, in order, are those of the reports' reproduction blocks, with the checks of Sections 4.6 and 4.7 and, last, the figure script:

```
# Acquisition and audit (Section 3; reports/data_audit.md)
.venv-lseg/bin/python scripts/acquire_lseg_fx.py --retrieval-date 2026-09-23
.venv/bin/python scripts/audit_fx_panel.py --retrieval-date 2026-09-23
.venv-lseg/bin/python scripts/acquire_lseg_fx.py --retrieval-date 2026-09-26
.venv/bin/python scripts/check_quote_revisions.py --new 2026-09-26 \
  --old 2026-09-23 2026-09-24
# Smiles, E1 and E4 (Sections 4.1 and 5.1; reports/smile_calibration.md, e1.md)
.venv/bin/python scripts/calibrate_smiles.py --start 2010-07-30 --end 2026-08-31
.venv/bin/python scripts/calibrate_smiles.py --contributor FN \
  --start 2007-01-31 --end 2026-08-31
.venv/bin/python scripts/estimate_e1.py
# Returns, E2, E3, E5 and moments (Sections 5.2-5.4; reports/stage4.md)
.venv/bin/python scripts/estimate_stage4.py
.venv/bin/python scripts/acquire_cboe_vx.py --retrieval-date 2026-09-24
.venv/bin/python scripts/estimate_e5.py
.venv/bin/python scripts/estimate_moments.py
# Robustness (Section 6; reports/robustness.md)
.venv/bin/python scripts/calibrate_vv.py
.venv/bin/python scripts/robustness.py
.venv-lseg/bin/python scripts/acquire_lseg_fx.py --retrieval-date 2026-09-24 \
  --skip-search --only <CCY>3M= <CCY>3MD= (nine currencies) \
  USD3MD= USD3MOIS= USDSROIS3M=
.venv/bin/python scripts/calibrate_smiles.py --tenor 3M \
  --extra-raw-root data/private/lseg/2026-09-24/raw
.venv/bin/python scripts/estimate_3m.py
# Post hoc analyses and validation (Sections 5.1, 5.2 and 4.6)
.venv/bin/python scripts/estimate_regime_attribution.py
.venv/bin/python scripts/validate_moments_models.py
.venv/bin/python scripts/estimate_identification.py
.venv-lseg/bin/python scripts/acquire_lseg_fx.py --retrieval-date 2026-09-27 \
  --skip-search --only <CCY>1MR5=FN <CCY>1MB5=FN (nine currencies)
.venv/bin/python scripts/estimate_wing_test.py
.venv/bin/python scripts/acquire_verdelhan.py --retrieval-date 2026-09-27
.venv/bin/python scripts/compare_verdelhan.py
.venv/bin/python scripts/check_bootstrap_arch.py
# Checks computed after the main results
# (Sections 3.2, 4.1, 4.6, 4.7, 5.2, 5.3, 5.4 and 6.1)
.venv/bin/python scripts/estimate_long_atm.py
.venv/bin/python scripts/estimate_design_checks.py
.venv/bin/python scripts/estimate_outstanding_items.py
.venv/bin/python scripts/describe_august_2024.py
.venv/bin/python scripts/check_reported_conditions.py
bash cpp/build.sh && .venv/bin/python scripts/benchmark_kernel.py
# Figures (Sections 5.1 to 5.3)
.venv/bin/python scripts/make_figures.py
```

The three-month forward points and rates come from the separate retrieval of 24 September 2026, which the three-month calibration reads; the header of `scripts/reproduce.sh` lists its 21 instruments. The kernel build needs a C++17 compiler. The robustness script stops if its base variant does not reproduce the E1 regime difference and the means of HML^U, HML^H and the skew term. The script `scripts/reproduce.sh` runs all of these steps except the acquisitions, the revision check, which needs the retrieval of 26 September, the kernel benchmark, which rewrites the timing block of `cpp/README.md`, and the figure script, in a new environment built from

`requirements.lock`. I ran it from a fresh clone on 3 October 2026: all tests passed (one optional check, which needs mpmath, was skipped), every step completed, and 23 of the 50 result files were byte-identical to mine; in the others every number agreed with mine to within 10^{-9} relative plus 10^{-12} absolute, apart from run times and a density diagnostic that differed by at most 4.2×10^{-10} and changed no flag. These numerical differences come from the linear-algebra library linked to the installed numpy. Three of the eight audit tables differed for reasons unrelated to the estimates: two list instruments added to the catalogue after the retrieval of 23 September, and one carries a corrected citation. The estimation code has not changed since that run; the figure script and the corrections of attribution of 10 October (Appendix C) postdate it.

Rerunning the acquisition, and hence the estimation, requires the reader's own LSEG Workspace licence, not mine; the acquisition script lets a reader with that licence rebuild the private layer. Raw tables, the selected month-end inputs, audit counts by instrument, per-currency month-level series and calibrated parameters are kept in `data/private/`, which Git ignores, and a repository test fails if anything under that folder, or a data file outside the published series in `data/public/`, is tracked. The Cboe settlement files are public but are stored privately because their redistribution terms are not established. The licence holder has confirmed that transformed data may be published and raw data may not (research log, 1 October 2026).

C Deviations, corrections and items not reported

The research log records every departure from the design with its date. Before any inference, the Newey–West lag, first set to the pilot truncation of Newey and West (1994), was replaced by the full automatic procedure, and the extended sample was clarified to require the calibration set only. Before the E1 results were recorded, a review led to New York calendar dates, provider days-to-maturity instead of a 30-day fallback, exclusion of currencies rather than whole months when inputs are missing, and a leg-by-leg splice test; each had a negligible effect on the primary estimates. Deposit rates are used for every non-USD currency, although the design allows OIS where available; the effect on the base-currency discount factor, the only channel through which rates enter E1, is bounded at about 0.1 per cent. Before the E2 and E3 results were recorded, two errors were corrected: the first E2 bootstrap resampled blocks of pairs, which reintroduced the sample's predictive relation into the null and inflated the bias about tenfold relative to Stambaugh's approximation; and the leg weights of θ_0 summed to two. After the first E2 and E3 results, and before the E5 results were recorded, return windows ending after the last spot observation were excluded; the E2 and E3 estimates changed only in the third significant figure, the Clark–West statistic from 1.640 to 1.639. The supplementary E1 analyses are post hoc. The open rules of the robustness grid were fixed after the primary results were recorded but before any variant was estimated, and the choices for the three-month tenor, vanna–volga smiles and moment intervals before those estimates were computed.

Five items named in the design or the data plan were computed only after the main results had been recorded, on 27 September 2026: the long at-the-money sample, the SOFR OIS check, the delta-method intervals, the check against the named-broker contributor and the closed-form checks of the moment code, together with the C++ kernel of the design's last stage. Their rules are stated in their scripts and in the research log. An independent review of the code preceded their reporting and led to four corrections, none of which changed an estimate: a confidence set of the long sample had been given a finite end that was an edge of the scanned grid; the long sample's rules had been described as fixed before any of its returns were seen, which is not true of its primary-window months; the kernel's bracket check could underflow for inputs far outside the project's range; and the test of a zero hedge cost, added after a first run, rested on a normal reference that the skewness of option payoffs makes unreliable. Every test of $\theta_{\mathrm{UB}} = 0$ and the studentised bootstrap in Section 5.3 are post hoc.

On 2 October 2026 I checked every item of the design and every commitment in the research

log against what had been computed and reported. Four items had not been computed: the E3 terms in each regime, which the design's statement of the contribution names; the stale-butterfly rule in the extended sample, which the log of 23 September applies there; the descriptive account of 5 August 2024; and the comparison of θ_{UB} with premium shares in the literature. I computed the first three under rules fixed in their scripts before they were run, and made the comparison (Sections 5.3, 5.4 and 6). The robustness grid, the three-month tenor and the secondary moment predictors had been run with 1,999 bootstrap draws instead of the design's 9,999, a departure the log had not recorded. I reran them with 9,999 draws, which changes no conclusion at the 5% level, and the tables report those results. A further check on 3 October found three more items: the payoff and volatility-level terms and θ_0 of the three-month variant, which the log of 24 September commits to E3 as at one month; the test-inversion sets of θ_{UB} in the robustness grid; and static-arbitrage checks of the daily smiles behind the account of 5 August 2024. I computed them (Sections 5.4 and 6). It also found a departure the log had not recorded: the design fixes initial training on the extended sample for the out-of-sample test of E2, but the robustness grid trains every variant from the primary start, a rule fixed before any variant was estimated and described in the log as one the design leaves open. The final clean-environment reproduction dates from 3 October (Appendix B).

On 9 and 10 October 2026 I checked every attribution in this paper, the theory notes and the code against its source, and searched for prior statements of each result I had claimed; I completed the resulting corrections on 15 and 16 October. The check withdrew my claim to the slope condition of R5(c), which is the inequality of Fukasawa (2012, Lemma 2.1) in two-sided form; narrowed my claims for R4 and R10 to the parts stated in Section 2; corrected the description of the hedge of Jurek (2014), whose option delta is itself hedged; added the related work cited in Sections 2 and 4 and in the introduction; and credited two routines that follow existing implementations step by step, the pairwise summation of NumPy in the C++ kernel and the block-length rule of the arch package. No estimate changed. The figures, and the observation in Section 5.2 that the out-of-sample statistic owes most of its size to forecasts of 2020, were added at the same time and are post hoc.

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The version of each source that I consulted, and what I consulted it for, are recorded in `references.md` in the repository; page, section, table and equation numbers in the text refer to those versions.

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